

Master's Thesis

Explain-Then-Act: A Novel LLM-Critic Framework for Safer Reinforcement Learning in Autonomous Driving

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List of Abbreviations

AI	Artificial Intelligence
ADS	Autonomous Driving Systems
AV	Autonomous Vehicles
DRL	Deep Reinforcement Learning
GAIL	Generative Adversarial Imitation Learning
GPS	Global Positioning System
IL	Imitation Learning
IMU	Inertial Measurement Unit
IRL	Inverse Reinforcement Learning
LLM	Large Language Model
PPO	Proximal Policy Optimization
TTC	Time-to-Collision
RL	Reinforcement Learning
XAI	Explainable Artificial Intelligence
XRL	Explainable Reinforcement Learning

Abstract

Autonomous driving systems increasingly rely on deep reinforcement learning for decision making in dynamic traffic environments, yet these models operate as black boxes and lack mechanisms for transparent, safety-oriented reasoning. This thesis proposes an Explain-Then-Act framework that integrates a large language model as a safety critic within a reinforcement learning based driving policy to enable pre-action evaluation and structured self-critique. Before execution, each action generated by a Proximal Policy Optimization agent is assessed using contextual traffic information, including ego state, surrounding vehicles, headway distance, and time-to-collision metrics. The language model classifies actions as safe, risky, or unsafe and provides a natural language explanation, triggering re-planning when necessary. A hybrid reward function incorporates both environmental feedback and safety critique signals to guide policy learning. Evaluation in a simulated highway environment demonstrates improved safety performance and more reliable behavior while maintaining competitive efficiency. The proposed framework advances explainable reinforcement learning by embedding interpretability and safety assessment directly into the decision-making process, contributing to more transparent and accountable autonomous driving systems.

1 Introduction

Mobility has been one of the key forces shaping human evolution and civilizations. It has influenced the development of economies and cultures throughout history. The construction of roads and other infrastructure, the growth of transportation-related knowledge, and the technological advances of the Industrial Revolution played central roles in this process. From early animal-based transportation to the invention of the wheel, the construction of paved roads, and the first motorized vehicle built by Karl Benz in 1886, each stage in the history of transportation transformed how people moved from one place to another and how they interacted with their environment. These milestones in mobility shifted toward a new horizon and led to the emergence of personalized transportation from shared public systems. By the twenty-first century, technological advancements such as Artificial Intelligence (AI), data-driven decision making, and sensor technologies reshaped the narrative of mobility and began new era.

The introduction of Autonomous Vehicles (AV) is a key milestone in this journey, which allows us to think of a vehicle operating without a human driver. Autonomous driving refers to operating a vehicle with partial or no human involvement by using hardware and software that control vehicle movements and make decisions based on real-time data from onboard sensors. According to the Society of Automotive Engineers (SAE) and globally available knowledge, the autonomy levels (as shown in Figure 1.1) of autonomous driving are divided into six levels (0 to 5) based on technological availability and vehicle characteristics, such as acceleration, steering, perception, and fallback capabilities (Automotive Engineers (SAE), 2021). The journey toward full automation or level 5 can not be achieved overnight. Today's autonomous driving systems reach level 3 and are available under specific conditions, and are progressing towards level 4. The final stage of full autonomy remains theoretical or testing in controlled environments. The evolution of autonomous driving is expected to reduce traffic congestion and enhance safety and accessibility. These outcomes depend on the system's capacity to make real-time decisions and manage complex driving tasks.

SAE Level	Name	Steering & Acceleration	Perception	Fallback	System capabilities
Human in charge of perception					
0	No Automation	Driver	Driver	Driver	None
1	Driver Assistance	Driver + System	Driver	Driver	Some driving modes
2	Partial Automation	System	Driver	Driver	Some driving modes
System in charge of perception					
3	Conditional Automation	System	System	Driver	Some driving modes
4	High Automation	System	System	System	Some driving modes
5	Full Automation	System	System	System	All driving modes

Figure 1.1: Autonomy Levels defined by Society of Automotive Engineers (SAE)
(Automotive Engineers (SAE), 2021)

1.1 Background of the Study

Traditional human-controlled driving, which we are familiar with, relies solely on the human driver and the decisions they make. The driver has full responsibility for detecting and interpreting road elements and other traffic participants, determining appropriate speeds, lane changes, maneuvers, and translating those decisions into physical actions such as braking, accelerating, or steering. In contrast, autonomous driving systems shift the majority of these responsibilities from the human driver to the machine, using various sensors, perception modules, and decision-making algorithms to perform tasks that previously required continuous human attention. This transition did not occur overnight but instead progressed through a series of stages, beginning with driver-assistance systems such as adaptive cruise control, lane keeping, and moving toward full autonomy, in which the system manages end-to-end driving with little or no human intervention.

The Autonomous Driving Systems (ADS) is the backbone of the current autonomous driving stack, processing raw data from the various vehicle-mounted sensors to accomplish safe and goal-oriented behavior. Modern stacks normally begin with perception, followed by prediction, planning, and control modules. Today’s ADS are moving away from traditional modular architectures toward end-to-end systems that generate trajectories directly from raw sensor data without explicitly separating the process into different modules. Such end-to-end systems have shown performance comparable to modular architectures in ADS (Chen et al., 2024).

Overview of the Autonomous Driving System (ADS) Pipeline

As illustrated in Figure 1.2, in a modular architecture for ADS, information flows from the sensing module to vehicle actuation through a structured framework. Each module passes critical inputs to subsequent stages while providing a simplified architecture for development and debugging.

A modular ADS begins with the sensing module, which includes multiple sensors mounted on

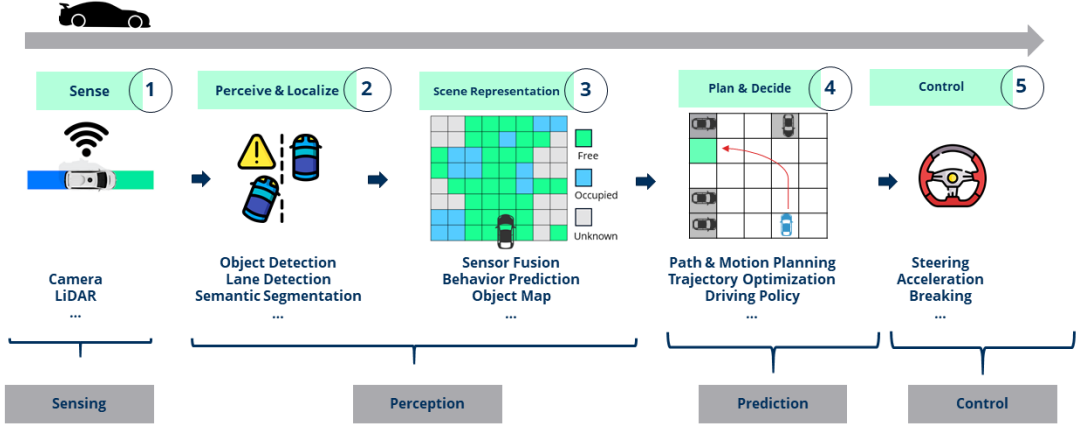


Figure 1.2: Key Components in the Modular-based Autonomous Driving Systems Pipeline

the vehicle, such as cameras, LiDAR, Radar, Global Positioning System (GPS), and Inertial Measurement Unit (IMU). These sensors provide essential as well as critical information about objects, including appearance, depth, surface characteristics, and motion. Through localization and mapping, the ADS further strengthens its capability in real-world environments by allowing the vehicle to determine its position within the surroundings. Scene representation allows the generation of a structured representation of the environment, often using grid-based formats, 3D point clouds from LiDAR sensors, and bird’s-eye-view representations of the environment, commonly referred to as scene representations (Mozaffari et al., 2022). This representation simplifies the environment by enabling the vehicle to identify free and occupied spaces through computer vision methods. In addition, combining perception and localization with scene representation supports object detection, object tracking, and lane detection, which are essential for a robust ADS. The output of this module is then passed to the planning and decision stage, which is often considered the core of ADS. The primary task at this stage is motion prediction, which estimates how the vehicle will interact with other road users such as pedestrians, cyclists, and other vehicles. Finally, in the control module is responsible for executing the trajectories generated by the planning module. It directly influences the ego-vehicle by regulating steering, throttle, and braking inputs to achieve accurate and stable vehicle motion.

Although the ADS pipeline, shown in Figure 1.2, presents a structured flow of information from sensing to actuation, the most consequential decisions are made within the planning and decision layer. This layer determines how the vehicle responds to dynamic environments and how it interacts with other road users, including pedestrians, cyclists, and motorists. It also responsible for selecting trajectories that balance safety and efficiency. Because these decisions directly influence vehicle behavior, a large body of research has focused on the algorithms that govern this layer, which is commonly referred to as driving policies.

1.2 Problem Statement

The field of research and development in autonomous driving has progressed steadily and promisingly over the past decade, with the help of advances in sensing technologies, supporting hardware, as well as modeling and simulation frameworks designed to imitate real-world driving conditions. The commercial platforms, such as Tesla Autopilot and Waymo Driver, are capable of performing selected driving tasks and continue to enhance themselves in areas concerning object detection, vehicle maneuvering, and perception. Despite these developments in the field, self-driving vehicles continue to show failures that raise concerns among the public as well as manufacturers and regulators regarding the robustness and reliability of the system.

Concerns about trust have been proven by several recent incidents. There have been several financial penalties imposed on the manufacturers following fatal accidents, large-scale recalls by companies, and several incidents involving traffic law offenses. These events have contributed to negative public perception and damaged confidence in the technology, leading many to question whether such driving systems are safe and reliable on public roads. Authorities also continue to question safety and compliance with basic traffic rules, since [AV](#) must operate on public roads alongside conventional vehicles while adhering to established regulations.

As a result, explainability in autonomous driving systems has become critical and essential, and it should be addressed in a structured and systematic manner. In traditional human-centric driving, drivers often establish a clear communication channel with passengers that conveys the driver's intended decision-making process and clarifies why a particular action is taken based on the surrounding context. Such explanations uplift the confidence level of the passengers and create trust in the whole driving process. Current research efforts have attempted to address the explainability issue in [ADS](#) through visualizations and post hoc explanations. However, these approaches have limited applicability in real-time and dynamic settings, where proactive explanations are important. Passengers and other responsible parties can monitor and diagnose system behavior more effectively when the internal decision-making process is understood. Further, regulatory authorities are required to evaluate and certify [AV](#) systems based on standards, criteria, and traffic laws, which further highlight the need for interpretable and transparent decision-making mechanisms.

This creates an important question about why modern autonomous driving systems are not sufficiently capable of explaining their own behavior. Today's [ADS](#) primarily rely on Deep Reinforcement Learning ([DRL](#)) as it gives promising performance in complex driving settings. The Reinforcement Learning ([RL](#)) based design enables the autonomous vehicle to interact as an agent with a simulated environment through a reward and penalty system. Yet although this system produces strong performance, which is obviously better than a set of manually defined actions, the decision-making process is buried deep inside the network. The internal features are

stored in latent representations, and their outputs do not disclose the rationale behind specific decisions. This is often described as the black-box problem, which limits the ability to understand the mechanism behind why a particular action was chosen in a given context. Further, these systems do not provide human-understandable justifications for their decisions. For decades, people have established a good communication channel with a human driver, and the absence of such explanations in autonomous driving, particularly in safety-critical situations, reduces public confidence and trust. In addition, these systems are designed to produce only outputs without any form of self-criticism regarding their internal decision process. Without having a feedback mechanism of this kind, the learning process does not reach its full capacity and may lead to more unsafe behaviors when deployed in real-world driving.

The challenges described so far open a broader gap in the current autonomous driving literature. While modern [ADS](#) demonstrate strong and effective performance in complex and dynamic environments, they often lack transparency in how decisions are made internally. The absence of such clear explanation brings challenges for passengers, engineers, and regulators alike, all of whom need understandable reasoning to assess system behavior and ensure safety. Existing post-hoc explanation approaches only partially address this issue, as they provide insight after decisions have already been executed. Such limitations highlight the need for a method that support explanation as part of the decision-making process itself.

Motivated by these limitations, this thesis work brings novel *Explain-Then-Act* framework for autonomous driving. It investigates the integration of a large language model as a safety critic within a reinforcement learning-based driving system. In the proposed framework, the language model acts as a reasoning component that evaluates planned actions and provides natural language explanations with potential safety concerns before execution in the environment. It also provides human-understandable explanations, which can improve public trust and confidence in autonomous driving systems. By combining [RL](#) with Large Language Model ([LLM](#)) reasoning, this work aims to improve transparency by enabling internal self-critique and to contribute to safer, more trustworthy, and more interpretable autonomous driving systems.

2 Literature Review

Driving policies can be implemented using several methodological frameworks, each reflecting a distinct perspective on how an autonomous vehicle reasons about its environment. Early ADS primarily relied on rule-based and optimization-driven approaches that used manually designed logic and deterministic decision structures. In contrast, recent progress in machine learning has enabled data-driven policies that learn decision-making behavior from demonstrations or through interaction with the environment. Comparing these two classes of approaches provides insight into how ADS trade off explainability, adaptability, and validity when operating in actual conditions.

2.1 Classical Rule-Based vs Learning-Based Driving Policies

Classical Rule-Based Driving Policies

Classical Rule-based systems make decisions using predefined rules and structured logic derived from expert knowledge, which map perceived environmental states to specific driving actions. Zhao et al. (2021) proposes a rule-based framework that applies if-then logic to perception outputs obtained from the traffic environment in order to generate driving behaviors. The system selects an appropriate driving strategy based on observed traffic conditions and recurring patterns. In addition, it incorporates probabilistic models and machine learning components to support the anticipation of upcoming situations and improve decision-making under uncertainty. In their review of knowledge-driven decision-making, Hu et al. (2025) describe rule-based systems as decision modules that use expert knowledge encoded in “if-then” logic to generate driving behaviors from perception inputs. These systems rely on predefined traffic rules and driving strategies, and typically rely on conditional logic and reasoning to determine maneuvers (e.g., lane choice) in response to the vehicle’s environment. The authors note that rule-based systems offer strong Interpretability and enable predictable, regulation-compliant responses in structured scenarios.

Learning-Based Driving Policies

The learning-based driving policies refer to control and decision-making systems that use data-driven approaches, mainly machine learning algorithms such as [RL](#) and [DRL](#) to learn how to drive the vehicle safely through interaction with an environment. Hu et al. (2025) examines the role of data-driven methods in autonomous driving decision-making and planning, highlighting learning-based policies such as Imitation Learning ([IL](#)), Inverse Reinforcement Learning ([IRL](#)), and [RL](#). The authors describe [IL](#) approaches including behavioural cloning, [IRL](#) and Generative Adversarial Imitation Learning ([GAIL](#)); behavioural cloning targets direct replication of expert actions but struggles in novel scenarios, [IRL](#) infers hidden reward structures enabling more generalizable policies, and [GAIL](#) combines adversarial training with imitation to boost robustness.

By considering both approaches, each presents distinct drawbacks, particularly when deployed in dynamic environments. Hu et al. (2025) describes that classical rule-based approaches have significant limitations, such as being overly conservative and tending toward large rule sets that increase computational load and reduce flexibility in complex or unfamiliar traffic situations. They emphasize that the accumulation of explicit rules may hamper scalability in dynamic environments such as urban driving. In contrast, learning-based policies suffer from a lack of transparency. These opaque models often achieve great performance, but due to their black-box nature, create challenges for safety assurance, debugging, and regulatory validation in real-time deployment (Hu et al., 2025). The authors argue that although learning-based policies offer adaptability to complex traffic environments, they also bring challenges including data hunger, issues and safety assurance concerns.

Therefore, learning-driven driving policies also require additional tools that provide clear and consistent explainability and structured critique capabilities for decision-making rationale and identification of unsafe and inconsistent behavior. Without such tools, neither rule-driven methods nor learning-driven methods are capable of meeting the overall requirements of safety, interpretability, and robustness that are necessary for autonomous driving systems.

2.2 Reinforcement Learning in Autonomous Driving

Autonomous vehicles operating on public roads must interact with a wide range of agents across diverse driving conditions that vary in traffic density, road structure & dynamics of road infrastructures, and regulatory constraints. Many established decision-making frameworks rely on extensive hand-crafted rules and hierarchical decision structures to manage these interactions. Although such methods can function well in controlled or moderately complex scenarios, their scalability is limited. When multiple vehicles must be considered jointly, the number of possible interactions increases rapidly, which leads to substantial computational demands in dense traffic. In addition, manually designed rules often struggle to account for rare or safety-critical

situations, which results in limited generalization ability and a high engineering burden. Reinforcement learning offers a solution because it allows an agent to acquire policies through repeated interaction with a simulated environment rather than relying on manually specified decision logic in real-world environment. [RL](#) has shown promise in tasks involving interaction between agents, cooperative and competitive behaviors, and long-horizon planning, which are all relevant to autonomous driving.

2.2.1 Overview on Reinforcement Learning

Reinforcement learning is a feedback-driven machine learning framework in which an agent learns to guide within an environment by selecting actions and observing the resulting outcomes (Connell and Sridhar Mahadevan, 1999). The agent receives positive feedback when it selects the right actions and penalties when it selects poor or wrong actions. Through repeated interaction with the environment, the agent gradually learn by maximizing the total amount of positive reinforcement it accumulates over time. Figure [2.1](#) demonstrates the key components of an [RL](#) system.

- **Environment** - The environment represents the external world that responds to the agent's actions. It is commonly modeled as a Markov decision process (MDP) consisting of a set of states s , a set of actions a , transition probabilities $P(s' | s, a)$, and a reward function $R(s, a, s')$.
- **Agent** - The agent is the decision maker that interacts with the environment by taking actions and receiving feedback. Its objective is to learn an optimal policy $\pi^*(a | s)$ that maximizes expected cumulative reward. Simple agents may rely on linear policies, while more advanced agents often use neural networks or other non-linear function approximators.
- **States** - A state represents the current situation of the agent within the environment. It provides the necessary information for selecting the next action and must be represented in a way that retains essential details while remaining computationally efficient.
- **Actions** - Actions define the decisions available to the agent at any given state.
- **Reward** - A reward provides feedback on the agent's previous action. The goal of the agent is to maximize the cumulative reward it expects to receive across future time steps.

At time step t , let $r(s_t, a_t)$ denote the immediate reward received after taking action a_t in state s_t . The expected cumulative discounted reward from time t is defined as

$$G = \left\{ \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \right\} \quad (2.1)$$

where $\gamma \in [0, 1]$ is the discount factor. The objective of [RL](#) is to find a policy $\pi^*(a | s)$ that maximizes G (Irshayyid et al., 2024).

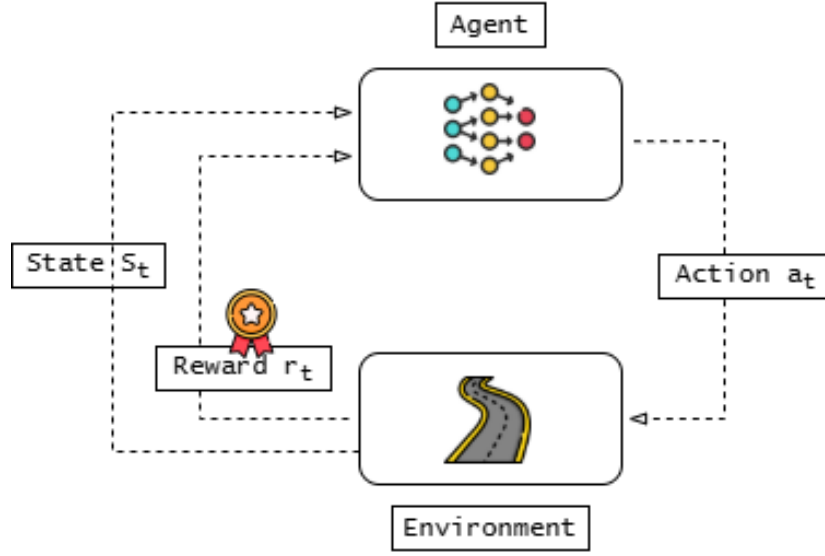


Figure 2.1: Key Components of a RL system

RL algorithms differ in how they estimate value functions or parameterize policies. Major categories include value-based methods such as Q-learning and SARSA, which estimate value functions and extract policies from them, policy gradient methods that directly optimize parameterized policies; actor-critic methods that learn both a policy and a value function and model-based methods that additionally learn a model of the MDP to support planning.

Over the past decade, reinforcement learning has become a popular technique among researchers because of its capability to achieve high-quality results across complex domains such as robotics, games, and, in particular, autonomous driving.

2.2.2 Taxonomy of Reinforcement Learning Algorithms

The reinforcement learning methods can be grouped into model-based and model-free approaches (Figure 2.2). Model-based methods attempt to learn a representation of the environment's dynamics and then use this learned model to compute an effective policy. In contrast, model-free methods do not attempt to model the environment. Instead, the agent learns an optimal policy directly through trial-and-error interaction, relying purely on observed states, actions, and rewards.

Model-Free Reinforcement Learning

In model-free reinforcement learning, two main approaches are used to represent and train agents: policy optimization and Q-learning

- **Policy Optimization** - Methods in this category define a policy directly as $\pi_{\theta}(a \mid s)$ and

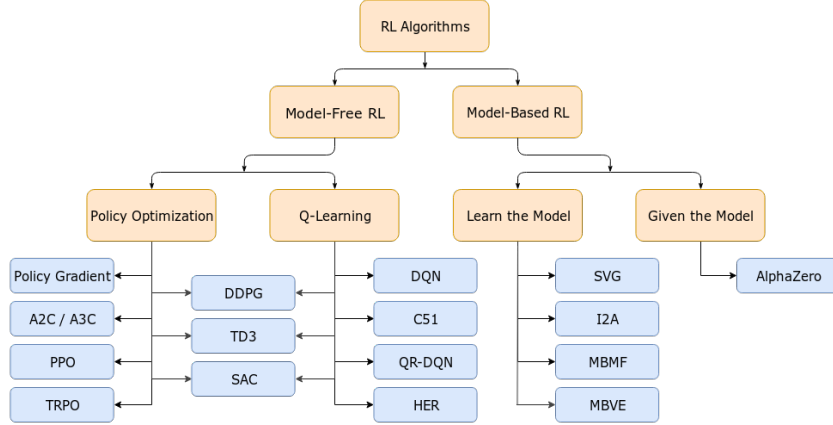


Figure 2.2: Taxonomy of RL Algorithm (Achiam, 2018)

adjust the parameters θ through gradient ascent on the performance objective $J(\pi_\theta)$, or by optimizing a local approximation of this objective. These methods are typically on-policy, meaning that each update relies solely on data gathered while following the most recent version of the policy. Policy optimization methods also commonly learn a value function approximator $V_\phi(s)$ that estimates the on-policy value $V_\pi(s)$. This value estimate provides essential guidance for determining how the policy should be updated (Achiam, 2018).

- **Q-Learning** - Methods in this group learn a function approximator $Q_\theta(s, a)$ for the optimal action-value function $Q^*(s, a)$. Their objective functions are generally derived from the Bellman equation. These algorithms are usually trained off-policy, which allows each update to use data collected at any time during training, regardless of the behavior the agent used when the data was generated. The policy is then recovered through the relationship between Q^* and π^* , so the agent selects actions according to

$$a(s) = \arg \max_a Q_\theta(s, a) \quad (2.2)$$

Model-based Reinforcement Learning

Model-based reinforcement learning is closely connected to ideas from control theory, where the objective is to plan through a learned or known dynamics function $f(s, a)$ in order to select optimal actions. It can be viewed as the part of [RL](#) in which the underlying physical laws of the environment are provided rather than discovered solely through interaction. Model-based methods often rely on stronger assumptions and approximations about the task structure, which can improve planning efficiency but also limit generality to environments that match those assumptions. There are two main approaches: learning the model or learn given the model (Achiam, 2018).

Deep Reinforcement Learning Algorithms

François-Lavet et al. (2018) presents [DRL](#) as the union of reinforcement learning and deep neural networks, enabling agents to operate in environments with high-dimensional inputs and learn policies without handcrafted features. They survey core algorithmic families including value-based (e.g., deep Q-networks), policy-based, and actor-critic methods and emphasize the challenge of generalization in [DRL](#) systems. The authors note that while early breakthroughs (such as in games) demonstrated the potential of [DRL](#), real-world applications require careful attention to state representation, stability of learning, and scalability.

Irshayid et al. (2024) discussed the application of [DRL](#) in autonomous vehicle control, especially tasks such as lane changing, ramp merging, and platoon coordination. Their review highlights how [DRL](#) frameworks must handle interactive dynamics among vehicles (state representations) and design reward functions that balance safety, efficiency and comfort in driving scenarios. They further point out that despite promising advances, significant gaps remain in applying [DRL](#) in real-traffic conditions, including robustness under uncertainty and interpretability of learned policies

2.3 The Black-box Nature in Autonomous Driving

The use of reinforcement learning, in particularly deep neural networks in autonomous driving introduces a significant black-box problem in which the internal reasoning of the system cannot be easily inspected or understood. Modern autonomous vehicles rely heavily on deep learning models for its decision making. These models operate through high-dimensional latent representations that are not interpretable to engineers or end users. As a result, it becomes difficult to determine why a vehicle selected a particular maneuver or whether its reasoning processes contain hidden failure modes. Researchers show that this opacity limits the ability to validate autonomous driving systems under all operating conditions because engineers cannot directly verify how the vehicle operates across rare or safety-critical scenarios (Kalra, 2016). The inability to open the black box complicates debugging, formal safety assurance, and post-incident analysis, which are essential for reliable deployment.

This lack of transparency directly make a doubt trust and societal acceptance among community. Users tend to trust systems when they can understand how decisions are made, especially in high-risk contexts such as driving. When autonomous driving systems cannot provide interpretable explanations, passengers and regulators have difficulty assessing reliability and accountability. Studies on [ADS](#) design emphasise that the opacity of deep learning reduces user confidence and raises concerns about hidden errors and bugs that may not be detected until a failure occurs. Broader work on interpretable machine learning further argues that black-box models are inappropriate for safety-critical domains because the inability to explain decisions undermines

responsible oversight and legal accountability (Doshi-Velez and Kim, 2017).

2.3.1 Understanding Black-Box Characteristics within the ADS Modular Pipeline

As discussed in Section 1.1, an ADS pipeline is typically organized into four main components: sensing, perception, planning & prediction, and control. Among these stages, the planning & decision phase is often described as the most opaque part of the system because it involves complex reasoning, forecasting future situations, and selecting high-level actions under uncertainty. These functions are difficult to model directly and frequently require advanced AI or learning-based methods. In contrast, perception and control tend to be more transparent because they rely on geometry and vehicle dynamics, which are well understood and observable in real-world scenarios.

Having identified that the planning module is the most opaque component of the ADS system, it is necessary to explore the causes of this opacity. Four related issues that together contribute to the black-box behavior of planning in state-of-the-art autonomous systems are described in the literature as follows: (1) the opacity of deep reinforcement learning policies, (2) the lack of human-understandable explanations for the results of the planning module, (3) the lack of internal critique capabilities in the learned policy, and (4) the lack of introspection of the actions produced.

Deep Reinforcement Learning policies are inherently opaque

Deep reinforcement learning policies are widely described in the literature as opaque computational entities that offer strong performance but limited insight into their internal decision processes. Neural policies rely on high-dimensional feature representations and nonlinear transformations that cannot be inspected in a direct or human-interpretable manner. These challenges become even more pronounced when policies incorporate sequential dependencies, value estimates, and reward-driven are combined. As noted by Cheng et al. (2025), deep reinforcement learning fundamentally relies on black box neural architectures. They argue that this architectural choice limits interpretability, undermines trust, and complicates deployment in safety-critical contexts such as autonomous driving. Their survey highlights that the root challenge is not specific to a particular algorithm but rather to the overall structure of deep reinforcement learning pipelines. This structure obscures how internal representations lead to a particular decision. Similarly, researchers in the field of Explainable Artificial Intelligence (XAI) also explain deep neural networks as opaque computational model whose internal mechanisms cannot be inspected in a straightforward way (Hassija et al., 2024). In reinforcement learning settings, this opacity is further amplified by the need to reason over time, incorporate stored value estimates, and account for long-term reward objectives, all of which interact in ways that make post-hoc interpretation particularly challenging.

Heuillet et al. (2021) reviews explainability methods for reinforcement learning and highlights that DRL-based models remain difficult to interpret, especially in situations where explainability is required. The authors state that existing models fail to provide clear explanations in applications that require accountability and understandable decision-making processes. This leads to a broader topic in XAI, where models based on deep neural networks learn from distributed internal features that do not correspond to how humans explain concepts. This highlights how difficult it is to trace downstream actions back to their underlying causes. Systematic reviews further highlight that RL agents often produce decisions without interpretable justification, which creates challenges in understanding policy behavior at both local and global levels. Wells and Bednarz (2021) further elaborates that RL agents function as black boxes, and many scholars have focused on explaining and opening this black box through visualizations, query-based explanations, and policy summarization. Such efforts provide only partial or state-specific insights and do not offer a comprehensive explanation of how decisions are made across different scenarios. Attempts to analyze or visualize these policies usually provide only local explanations that reveal why the agent acted in one specific state, but do not provide a global understanding of the context in which the policy behaves across all scenarios. This represents a major gap in current approaches. Although these methods can achieve strong performance, their reasoning processes remain hidden and function as a black box from an explainability perspective.

Lack of Human Understandable Justifications

Research on explainable reinforcement learning consistently reports that standard RL models lack human-understandable justifications for their decisions, which creates practical and scientific limitations when users must interpret agent behavior. Heuillet et al. (2021) emphasizes that RL agents often function as black boxes in critical settings where clear justification is required, and they note that users frequently need straightforward answers about why an agent selected a specific action in a given state. This concern aligns with the work of Madumal et al. (2020), who argue that causal explanations of the form “A enables B and B causes C” improve communication between humans and agents and their results show that such causal chains support better outcomes in human-agent evaluations, which indicates that ordinary model-free RL methods do not provide these forms of causal reasoning without additional structure. Lin et al. (2021) adds that explaining why an agent favors one action over another calls for architectures that explicitly support explanation, which confirms that conventional RL pipelines lack built-in mechanisms for producing human-readable reasoning traces.

No Mechanism for Internal Critique or Feedback

Research on explainable reinforcement learning consistently shows that standard RL agents do not include mechanisms for internal critique or feedback, which limits their ability to monitor or analyze their own reasoning. Qing et al. (2025) state that classical RL frameworks prioritize the optimization of decision-making performance while leaving the internal decision-making logic

unexamined, which directly indicates the absence of built-in self-assessment. Wells and Bednarz (2021) observe that most Explainable Reinforcement Learning (XRL) methods concentrate on producing explanations at the policy or state level because agents otherwise provide no information about their behavior apart from observable actions and rewards, reinforcing the point that RL agents do not generate internal evaluations of their choices. Sequeira and Gervasio (2020) present a framework in which agents inspect their internal states for self reflection, and this contrast shows that typical systems do not possess such introspective features without additional design interventions. Hsu et al. (2024) describes recent attempts to introduce a safety critic that functions as a risk monitor, which further illustrates that feedback or critique is usually implemented as an external component rather than an inherent part of the agent’s planning process.

Actions are Non-introspectable or Non-interpretable

Introspectability refers to a system’s capacity to reveal the internal factors that shaped its decision, such as which inputs or latent features influenced an action. Interpretability refers to the degree to which humans can understand and verify the decision logic. In the autonomous driving literature, both concepts appear as recurring limitations in data-driven planning and reinforcement learning systems. Bai (2024) notes that reinforcement learning for autonomous vehicles continues to suffer from weak insight into why a specific action is chosen, especially in complex driving scenarios. Paleja et al. (2023) similarly report that gradient-based RL policies achieve strong control performance but lack interpretability, which limits their deployment in safety-critical and regulated environments such as autonomous driving. These findings indicate that many learned planning modules output an action without producing a traceable explanation of the reasoning path or of the dominant features that influenced the policy.

Interpretability challenges are also emphasized in broader studies of decision-making for autonomous vehicles. The survey on Explainable Artificial Intelligence for Autonomous stresses that current intelligent decision processes are generally not understandable to humans and this gap undermines social acceptance (Atakishiyev et al., 2024). Wang et al. (2021) further highlight that high-dimensional sensory input and limited semantic state representations hinder interpretability during highway on-ramp decision-making. In these cases, a system may compute a valid trajectory or manoeuvre, yet engineers, auditors or regulators cannot easily confirm whether the underlying reasoning aligns with safety goals or legal constraints. In one recent work by Kuhn et al. (2020) the authors treat the autonomous vehicle as a black box and train a failure-prediction model by using disengagement and successful driving sequences; the model aims to learn patterns of when the system cannot cope rather than revealing how the system produced its action. This further illustrates that the actions of many planning modules are non-introspective in the sense that they do not explain “why” or “how” they arrived at that choice, merely that they did or did not succeed. Together, these works show that the inability to introspect or interpret the

actions of modern neural planning modules remains a central barrier to trustworthy autonomous driving.

2.4 Explainable Reinforcement Learning (XRL)

[XRL](#) extends the broader goals of Explainable Artificial Intelligence by focusing on how agents that learn through trial and error can justify their policies and actions to human users. [XAI](#) seeks to make model behaviour transparent, interpretable and accountable, and Ali et al. (2023) define it as a set of methods that enable users to understand and trust model outputs. The need for interpretability is essential in reinforcement learning because policies often emerge from complex interactions within high-dimensional environments, which can lead to opaque decision processes similar to the black-box behaviour noted in surveys on autonomous driving systems (Ali et al., 2023). In safety-critical contexts, this lack of transparency creates challenges for trust, validation, and regulatory acceptance. As a result, [XRL](#) aims to produce explanations that clarify state-action relationships, reveal what the agent has learned and allow domain experts such as engineers & regulators to evaluate whether the learned behaviour aligns with safety or performance requirements.

Within the autonomous driving context, [XRL](#) is positioned as a mechanism to address concerns about real-time decision making that is difficult for humans to interpret. Atakishiyev et al. (2024) describe that the black-box nature of autonomous decision policies contributes to societal concerns about vehicle safety. Omeiza et al. (2022) further argue that autonomous vehicles should be able to explain what they have recognized, executed and plan to do, since transparency is closely tied to social acceptance and regulatory compliance. These perspectives motivate the use of [XRL](#) methods that generate human-understandable rationales for actions such as braking, lane changes or obstacle avoidance. Such explanations support debugging and post-incident assessment, which aligns with the broader view that explanation is essential in environments where accountability and trust are central requirements (Yeong et al., 2025). In this setting, [XRL](#) contributes by linking the internal reasoning of reinforcement learning agents to interpretable signals that can be examined by engineers, auditors and regulators.

2.4.1 Approaches to explainability in RL

In the taxonomy of [XAI](#), two approaches are defined based on when the information is extracted: intrinsic methods and post-hoc methods

Post-hoc explanations

In the context of [XRL](#), post-hoc approaches refer to techniques that are applied after a policy or model has been learned, with the aim of providing explanations for the behaviour of a possibly opaque model. According to Puiutta and Veith (2020), post-hoc interpretability is achieved

by analysing a trained [RL](#) model or its outputs (for example via surrogate models or feature-importance analyses) rather than by constructing an inherently transparent model. They note that post-hoc methods dominate current [XRL](#) research, since it is often more feasible to simplify or approximate an existing high-performing policy than to build one that is intrinsically simple from the scratch. However, this approach carries limitations: because the explanation is derived after the fact it may not faithfully capture the full decision process of the original model, and the performance-interpretability trade-off still holds such that explanations may either compromise fidelity (how well the predictions of the mimic model match those from the mimicked mode) or fail to provide meaningful insight (Puiutta and Veith, 2020). Additionally, Puiutta and Veith argue that many [XRL](#) methods offering post-hoc explanations neglect the human user in their design, omitting evaluation of how non-expert users understand or trust the resulting explanations.

Intrinsic explainability

Intrinsic explainability (or interpretability) refers to models that are designed from the outset to be transparent and understandable, rather than relying on a second layer of explanation applied later. In the [XRL](#) taxonomy, an intrinsically interpretable [RL](#) model is one where the structure of the policy or value function is constrained so that human users can more readily understand how decisions are reached (e.g., via decision trees, rule-based policies, or modular human-readable skills). The advantage is that such models avoid the surrogate-approximation layer and potentially offer higher fidelity between explanation and actual model behaviour. Nonetheless, Puiutta and Veith (2020) highlighted that intrinsic models often sacrifice performance for interpretability, and that in the domain of [RL](#), especially [DRL](#), the complexity of tasks. As a result, relatively few [XRL](#) works adopt intrinsic explainability compared to post-hoc, and the trade-off means that adoption of intrinsic methods may be limited in high-stakes or complex [RL](#) settings.

While explainable reinforcement learning has introduced valuable methods for improving transparency, most existing approaches remain constrained by their reliance on post hoc analysis or simplified intrinsic model structures. Post-hoc explanations often approximate policy behavior after decisions have already been executed, which limits their usefulness in safety-critical scenarios where errors must be identified and mitigated before action selection. Intrinsic explainability methods, although more faithful to the underlying policy, frequently impose structural constraints that reduce performance in complex and dynamic driving environments. As a result, current [XRL](#) techniques offer limited support for real-time safety assessment, decision justification, and proactive error prevention. These limitations motivate the exploration of alternative explainability mechanisms that operate prior to action execution and are capable of reasoning about safety and intent as part of the decision-making process, rather than explaining outcomes only after they occur.

2.5 Large Language Models (LLMs) for Reasoning and Explanation

2.5.1 Overview of Large Language Models

LLMs are deep learning systems trained on large corpora of text to predict or generate language from context and thereby capture patterns of syntax, semantics, and discourse in broad domains (Zhao et al., 2025). These models typically leverage the transformer architecture, are scaled to billions of parameters and process vast amounts of textual data to function as general sequence predictors (Kumar, 2024). Once trained, LLMs can be adapted via prompting or fine-tuning to tasks such as text generation, summarization, translation or question & answering. The applications have covered across sectors including healthcare, finance, education and autonomous systems. One specific application area involves integrating LLMs with vision models to support open-world reasoning in autonomous driving systems (Yang et al., 2024). Given their wide applicability and flexibility, LLMs are increasingly central to research and industry efforts in natural language processing and allied domains.

2.5.2 LLM as a Tool for High-level Reasoning

Large language models have been investigated as tools for high-level reasoning in natural language settings, with recent work examining how prompting strategies shape multi-step inference and decision support. Survey research has classified techniques that influence reasoning quality, including structured prompting methods that guide models through intermediate steps (Huang and Chang, 2023a). Chain of thought prompting in particular provides stepwise rationales that help models externalize intermediate computations, which enables more stable reasoning on tasks that require multi-hop inference. Method-oriented work has shown that providing explicit intermediate steps improves performance on mathematical, symbolic, and commonsense benchmarks (Huang and Chang, 2023b). Further research has explored search-based procedures in which models generate and compare alternative reasoning paths to support more deliberate problem solving, including extensions that apply chain-of-thought style reasoning to branching search processes (Yang et al., 2024). Together, these studies provide evidence that LLMs can be configured as reasoning systems when combined with structured guidance and evaluation.

2.5.3 LLMs for Explainability, Transparency and as Safety Critics

LLMs have been increasingly used to support explainability and transparency by expressing reasoning in natural language. The *Inner Monologue* framework shows that verbalizing internal thought processes during sequential decision making can make agent behavior more interpretable, particularly for embodied systems that operate over long horizons. Huang et al., 2022 describes how planning and self-reflection expressed in language allow users to observe intermediate goals and constraints that guide action selection. Relatedly, *Reflexion* demonstrates that language agents can generate explicit reflections after each action, which are then reused

to improve subsequent decisions. This framework highlights that reinforcing the agent through linguistic feedback, rather than updating network weights, can influence the learning process more quickly and efficiently. Shinn et al., 2023 show these self-generated natural language reflections function as a form of verbal reinforcement learning, enabling clearer inspection of why an agent changes its behavior over time. Beyond action-level reasoning, *DriveLM* illustrates that scene-level explanations can help users understand autonomous driving decisions by summarizing relevant environmental factors and planned responses (Sima et al., 2025). In addition, Azaria and Mitchell, 2023 demonstrate that LLMs encode latent reasoning cues within their internal activations, which can be probed to reveal whether a model knows when it is lying. This result supports the claim that LLMs provide interpretability signals that are not available in black-box deep reinforcement learning policies. Large Language Models have also been proposed as safety critics that monitor, constrain, and critique agent behavior. *SayCan* shows how high-level language reasoning can be grounded in robotic domain so that action choices respect physical constraints defined by the environment. Ahn et al., 2022 demonstrate that separating language-based reasoning from low-level control supports safer execution by limiting actions to those that are feasible. Building on this idea, safety-guided robotics with large language models uses LLMs to evaluate planned actions against safety guidelines expressed in natural language.

Large language models are increasingly studied as tools for explainability, transparency, and safety critique in autonomous driving systems when they are positioned as interpretable reasoning layers rather than direct control modules. Chen et al., 2023 present an architecture in which object level, numeric vector representations from perception are aligned with language tokens and fused with a pre-trained LLM to generate question answering style explanations of driving scenes, demonstrating that structured kinematic and semantic context can be converted into concise textual prompts that the model can reliably interpret. Further, Cui et al., 2025 survey a wide range of LLM applications in autonomous driving and provide a taxonomy of LLM roles that includes scene explanation, planning support, and safety monitoring, which helps frame the use of LLMs in a pre-action, introspective role focused on explanation and critique rather than post hoc narration alone. The survey also discusses evaluation metrics for LLM enabled driving modules, offering guidance on how explanation quality and safety related reasoning can be assessed in a principled manner. Together, these works support the view that LLMs can act as safety critics and transparency mechanisms by reasoning over structured driving context and articulating interpretable justifications for system behavior.

2.6 LLM-RL Integration for Safer and More Transparent Decision-Making

Large language model and reinforcement learning integration has been explored as a structured approach for improving safety and transparency in sequential decision-making systems, with

autonomous driving serving as a primary experimental domain. Xu et al., 2025 propose *TeLL-Drive*, which enhances autonomous driving by using a teacher large language model to deliver high-level strategies and risk-aware guidance to a student deep reinforcement learning policy. The framework incorporates language model chain-of-thought outputs into policy learning through attention-based fusion and scenario retrieval. This design allows abstract reasoning and contextual safety knowledge to influence low-level control learning in a traceable manner. Similarly, Pang et al., 2024 present *LGDRRL*, which employs an LLM-derived driving expert to guide deep reinforcement learning through intermediate reward generation and exploration shaping. The emphasis on intermediate rewards highlights how semantic safety assessments from language models can be translated into learning signals. These approaches improve transparency by making safety-related reasoning observable in the form of guidance, rewards, or retrieved scenarios.

A different integration strategy is presented in Shao et al., 2023 through *LMDrive*. It integrates a large language model into a closed-loop, end-to-end driving system that processes multiple input types and directly issues control signals for decision execution. The design exposes trade-offs related to latency and reliability when language models operate within the control loop. In contrast, critic-based or teacher-based methods do not directly influence on control rather contribute to shaping policy experiences indirectly. Within *TeLL-Drive*, *LGDRRL*, and *LMDrive*, the literature shows that both safety and explainability in the decision-making process can be supported when language models provide explicit evaluation, guidance, or reward shaping. These roles maintain the stability of reinforcement learning while making safety considerations more interpretable during training and evaluation.

3 Methodology

3.1 Problem Formulation

The decision-making task for autonomous driving can be formalized as a Partially Observable Markov Decision Process (POMDP). It can be expressed as a tuple $\langle S, A, O, T, R, \gamma \rangle$, where S denotes the set of environmental states, A represents the action space, O defines the observation space, T is the state transition function, R is the reward function, and γ is the discount factor ($\gamma \in [0, 1]$), which balances the weight on immediate and future rewards. The goal of the agent is to learn a policy π that maximizes the expected discounted return (Xu et al., 2025):

$$\max_{\pi} J(\pi) = \arg \max_{\pi} \mathbb{E}_{(s_t, a_t) \sim \rho_{\pi}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (3.1)$$

A. Observation Space

At each time step t , the agent receives an observation $o_t \in O$ that contains two main components.

$$o_t = [M_t; F_{\text{ego}, t}] \quad (3.2)$$

The first component is a matrix $M_t \in \mathbb{R}^{F_k \times N}$, which summarizes the state of N surrounding vehicles. Each column of M_t corresponds to one neighboring vehicle and is represented by a feature vector

$$F_k = [x_k, y_k, v_{x_k}, v_{y_k}, \cos(\theta_k), \sin(\theta_k)], \quad (3.3)$$

where (x_k, y_k) describes the k -th vehicle's position, (v_{x_k}, v_{y_k}) describes its velocity, and the hyperbolic cosine and sine of θ_k provide the representation of the vehicle's orientation. This feature design helps the model capture both spatial relationships and motion patterns in a numerically stable way.

The second component of o_t records the current state of the ego vehicle. This typically includes its position, velocity, heading, and any additional quantities needed for decision making. By combining the surrounding-vehicle matrix with the ego-vehicle state, the agent receives a compact

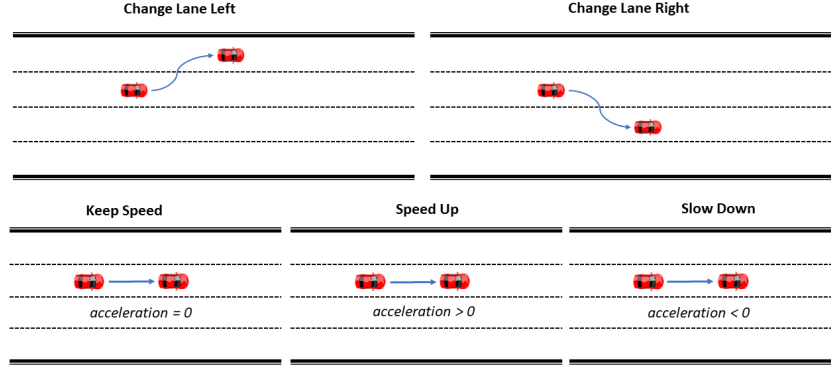


Figure 3.1: Action Space

but in detail description of the traffic scene. The combined observation offers enough information to capture local interactions, anticipate short-term motion, and provide reliable decision making in dynamic driving environments.

$$F_{\text{ego},t} = [x_t^{\text{ego}}, y_t^{\text{ego}}, v_{x,t}^{\text{ego}}, v_{y,t}^{\text{ego}}, \cos(\theta_t^{\text{ego}}), \sin(\theta_t^{\text{ego}})] \quad (3.4)$$

B. Action Space

The action space consists of a set of discrete driving decisions available to the agent. These actions are `keep_speed`, `speed_up`, `slow_down`, `change_lane_right`, and `change_lane_left`. These five actions represent the fundamental maneuvers required for longitudinal and lateral control in a multi-lane driving environment.

$$A = \{\text{keep_speed}, \text{slow_down}, \text{speed_up}, \text{change_lane_left}, \text{change_lane_right}\} \quad (3.5)$$

3.2 Model Frameworks

The proposed **Explain–Then–Act** framework for safer and more interpretable autonomous driving is an enhanced version of classical reinforcement learning.

In classical [RL](#) setup, an agent observes the current state s_t , selects an action through its policy $a_t \sim \pi_\theta(a \mid s_t)$, and executes that action immediately. This direct mapping from policy output to environment interaction can be risky in particularly safety-critical domains like autonomous driving, because the agent has no mechanism for self-reflection on the consequences of its chosen action before interacting with the given environment. To address this limitation, *Explain-Then-Act* framework has been proposed, which integrates a Large Language Model as a safety critic

that evaluates the action chosen by the agent within its context and provides feedback on potential risks before the action is executed in the environment. Based on the critique, the agent select another action a_t^* . This action is evaluated again by the safety critic, and the process continues until the action is classified as safe or the system reaches the maximum number of allowed re-planning attempts. With that the process can be summarized as:

$$s_t \rightarrow a_t^{(0)} \rightarrow \text{safety critique} = c_t = f_{\text{LLM}}(s_t, a_t^{(0)}) \rightarrow a_t^* \rightarrow \text{environment setup} \quad (3.6)$$

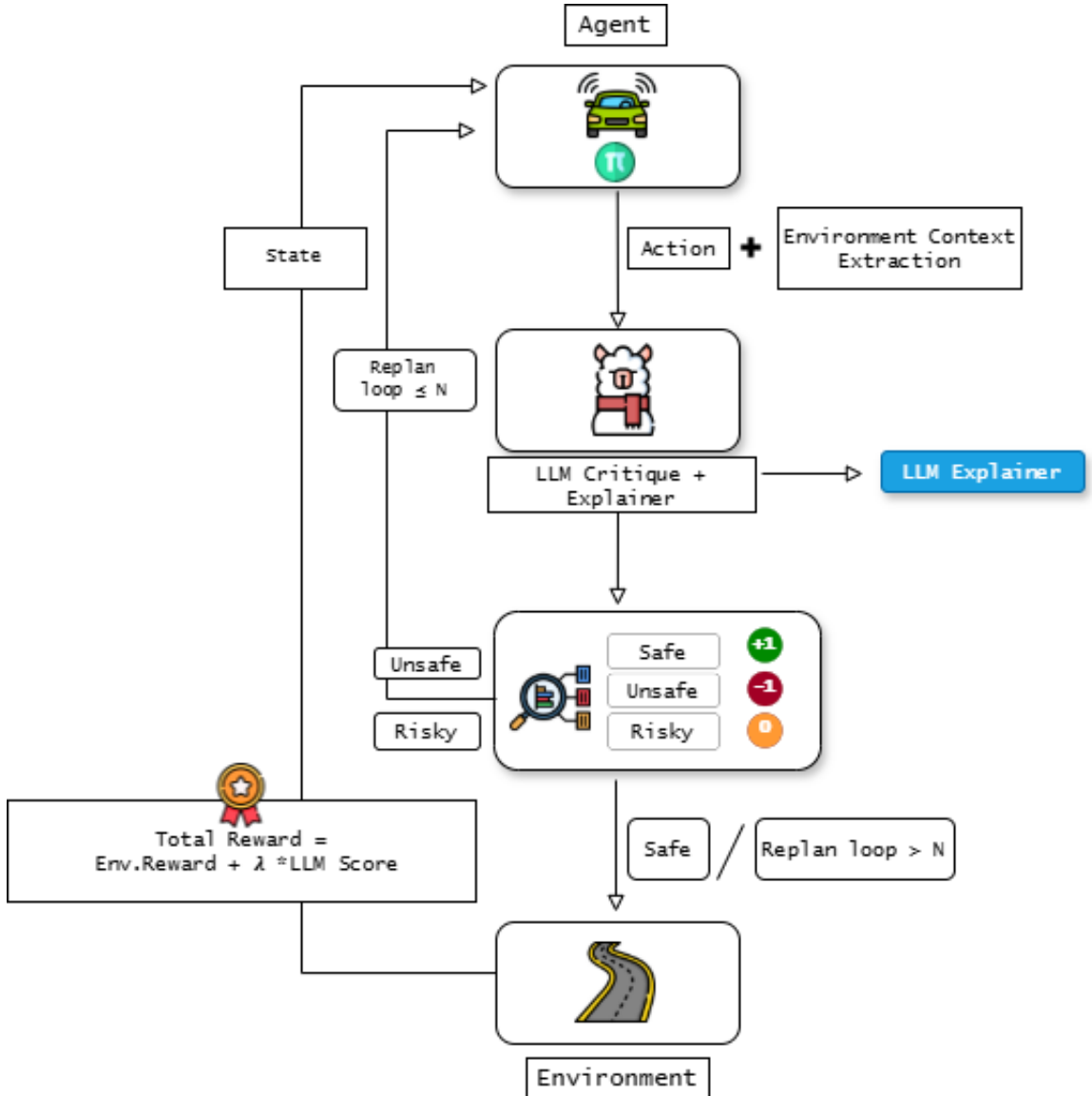


Figure 3.2: Overall Conceptual Framework for Explain-Then-Act

Algorithm 1 Explain–Then–Act

```

    Policy  $\pi_\theta$ 
    LLM critic  $f_{\text{LLM}}$ 
Require: Environment Env
    Coefficient  $\lambda$ 
    Maximum re-planning steps  $K_{\text{max}}$ 
1: for each timestep  $t$  do
2:   Observe state  $s_t$ 
3:   Propose initial action  $a_t^{(0)} \sim \pi_\theta(a \mid s_t)$ 
4:   for  $k = 0$  to  $K_{\text{max}}$  do
5:     Generate critique  $c_t^{(k)} = f_{\text{LLM}}(s_t, a_t^{(k)})$ 
6:     Compute score  $\ell(c_t^{(k)})$ 
7:     if  $\ell(c_t^{(k)}) = +1$  then
8:        $a_t^* \leftarrow a_t^{(k)}$  ▷ Safe action found
9:       break
10:    else
11:       $a_t^{(k+1)} \sim \pi_\theta(a \mid s_t)$  ▷ Replan
12:    end if
13:  end for
14:  Execute action  $a_t^*$ 
15:  Obtain environment reward  $r_{\text{env}}$ 
16:   $r_{\text{total}} = r_{\text{env}} + \lambda \ell(c_t)$ 
17:  Store transition in PPO buffer
18: end for
19: Update  $\theta$  using PPO

```

3.2.1 LLM Safety Critique Module

The safety critique module is designed with a large language model that evaluates the action proposed by the reinforcement learning policy. The primary goal of this component is to provide a qualitative judgment of safety before the agent commits to an action. To achieve this, the [LLM](#) receives a structured prompt that provides the current driving context and the selected action. The prompt classifies the action along with an appropriate explanation in natural language. Let’s say at timestep t , the [LLM](#) receives the prompt generated from the mapping function [3.7](#), where $\Phi(\cdot)$ represents the conversion from environment state S_t and proposed action $a_t^{(k)}$ into a textual description.

$$c_t = \text{LLM}(\Phi(s_t, a_t^{(k)})) \quad (3.7)$$

The c_t is the [LLM](#) critique classification [3.8](#) which provides the level of risk associated with the proposed action in a given context. Further numerical mapping ([3.9](#)) is then applied to convert these qualitative classification as simple score that has been incorporated in the reward function.

$$c_t \in \{\text{Safe}, \text{Risky}, \text{Unsafe}\} \quad (3.8)$$

$$\ell(c_t) = \begin{cases} +1, & \text{if } c_t = \text{safe}, \\ 0, & \text{if } c_t = \text{risky}, \\ -1, & \text{if } c_t = \text{unsafe} \end{cases} \quad (3.9)$$

Prompt Construction

A key requirement of the safety critique module is the design of the strong prompt provided to the LLM. The primary objective is to design a prompt with enough information about the driving context together with the proposed action, while keeping the prompt structured and readable. The prompt has several components, each representing a different aspect of the driving environment.

A. Ego Vehicle

As shown in Figure 3.3, red colour vehicle represents the ego-vehicle. The ego vehicle's lane index, position, speed, and heading angle are recorded for the prompt. These variables allow LLM to evaluate whether the proposed maneuver is feasible within the given context and road constraints.

B. Surrounding Vehicle

Figure 3.3 shows all surrounding vehicles in yellow color. The purpose of considering the surrounding vehicles is to understand the broader context and how it may influence on ego-vehicle. Therefore similar to ego-vehicle, lane index, position, speed and heading angle of closet vehicle in each lane are recorded for the prompt creation.

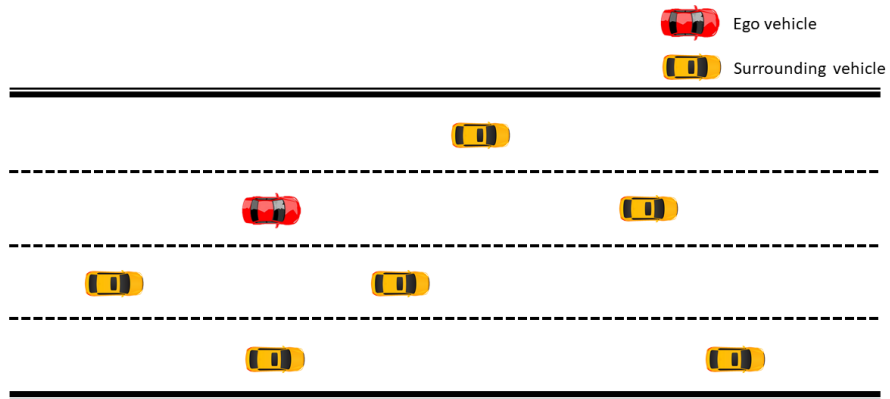


Figure 3.3: Environment Setup

In addition to the recorded features, the headway distance and Time-to-Collision (**TTC**) are calculated to strengthen the prompt. These two numerical values are strong indicators and explain how quickly a collision may occur if the ego-vehicle doesn't not change either its speed or its orientation. To maintain the simplicity, the longitudinal spacing is calculated between ego vehicle and closet vehicle in each lane. **TTC** estimates how long it will take the ego-vehicle to collide with front vehicle if both maintain their current velocities. For a vehicle directly ahead of the ego vehicle, **TTC** is defined as 3.10, where $\Delta v_k = v_{\text{ego}} - v_k$. This ensures that **TTC** is only finite when the ego vehicle is moving faster than the vehicle ahead. When $\Delta v_k \leq 0$, the ego vehicle is not reaching the front vehicle, and therefore a collision is not possible based on current velocities.

$$\text{TTC}_k = \begin{cases} \frac{d_k^{\text{headway}}}{\Delta v_k}, & \text{if } \Delta v_k > 0, \\ +\infty, & \text{otherwise} \end{cases} \quad (3.10)$$

3.2.2 LLM Guided Re-planning

One of the key module in the *Explain-Then-Act* framework is LLM-guided re-planning unit, which influences the final action before it is executed in the environment. Once the **LLM** critique classify the proposed action $a_t^{(k)}$ as **unsafe** or presenting a non-negligible level of risk, the system initiate the re-planning cycle. The condition for the system to initiate a re-plan can be expressed as

$$\ell(c_t^{(k)}) \leq 0 \Rightarrow \text{replan} \quad (3.11)$$

The re-planning process is designed to allow the agent to generate multiple candidate actions before interacting with the environment. When re-planning is triggered, the policy proposes a new action $a_t^{(k+1)}$ using the same state information 3.12. The new action is then evaluated by the LLM-critique and classified according to its level of risk. The process continues until the proposed action is classified as **safe** or until it reaches the maximum number of allowed re-planning attempts $k \leq K_{\text{max}} = n - \text{replans}$, which prevents the agent from entering an infinite re-planning loop. This upper bound is chosen to balance computational cost with the need for safety assurance.

$$a_t^{(k+1)} \sim \pi_{\theta}(a \mid s_t) \quad (3.12)$$

once the re-planning cycle terminates, the final proposed action $a_t^* = a_t^{(k)}$ may correspond to the safe option in the context or the least unsafe option within the allowed re-attempts.

3.2.3 Hybrid Reward Function

In traditional **RL** setup, the reward function only depends on environment's response to the agent's executed action. This can be expressed as $r(s_t, a_t)$, which capture the objective related

to performance such as avoiding collisions.

The proposed *Explain-Then-Act* framework introduces a hybrid reward function that not only weights the agent interaction with the environment but also incorporates the reward signals generated by the safety-critic. As shown in the Equation 3.9, the scalar safety score produced by the safety-critic and controlling parameter λ is used to compute new total reward at time step t for the refined action a_t^* . Then the total reward r_t^{total} can be presented as ;

$$r_t^{\text{total}} = r(s_t, a_t^*) + \lambda \ell(c_t) \quad (3.13)$$

The parameter λ ($0 \leq \lambda \leq 1$) defines the level of influence that the LLM-critique module has on the reward function. A larger value, close to 1, increases the contribution from safety critic, whereas a value close to zero keeps the reward closer to the original environment.

3.2.4 PPO-Driven Policy Optimization

The Proximal Policy Optimization (PPO) is used as the primary reinforcement learning algorithm due to its stability and reliable performance in a safety critical scenario like autonomous driving. It improves the policy by maximizing a clipped surrogate objective defined as

$$L_{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right]$$

where

$$r_t(\theta) = \frac{\pi_\theta(a_t | s_t)}{\pi_{\theta_{\text{old}}}(a_t | s_t)}$$

is the probability ratio and $\hat{A}_t$ is advantage estimator at timestep t (Schulman et al., 2017). The clipped objective in the PPO prevents the policy making excessively large updates and it avoid instability during the training. In a safety critical domain, this mechanism avoid abrupt policy changes that leads to unsafe behaviors.

In the *Explain-Then-Act* framework, PPO is implemented with additional reward component, that we discussed in the section 3.2.3. The objective function discussed in the Equation 2.1 is extended with the new total reward equation 3.13. It can be summarized to a equation as;

$$J(\pi_\theta) = \mathbb{E} \left[\sum_{t=0}^T \gamma^t (r(s_t, a_t^*) + \lambda \ell(c_t)) \right]$$

The proposed methodology attempt to address the key research gap identified in the current literature. As highlighted in the section 2.2.2, The overall *Explain-Then-Act* framework is designed to address the explainability limitations of traditional RL-based training and the opacity of deep reinforcement learning policies. In addition, LLM-based architectures provide human-understandable explanations by offering causal reasoning in natural language, with explanations

grounded in the driving context. To address the absence of internal critique or feedback, which is a major contributor to the black-box nature of current ADS, an LLM-based critic with re-planning units is introduced, as explained in Section 3.2.1. This allows the system to evaluate and revise actions before they are executed in the simulated environment. Finally, the framework provides clear explanations that describe why an action was selected and how the decision was reached. Such explanations help reduce barriers to trustworthy autonomous driving and narrow the gap between human-centric and autonomous driving systems.

3.3 Safety Properties and Design Rationale

Traditional safety mechanisms used in reinforcement learning-based autonomous driving systems tend to rely on hard constraints to restrict actions and overrule policy outputs when certain specified conditions are violated. Although such constraints are useful for ensuring adherence to certain safety rules, these constraints tend to be inherently brittle when operating in complex and dynamic real-world settings such as those encountered in autonomous driving systems. Hard constraints require exhaustive enumeration of unsafe situations, which is not very feasible when operating in open-world settings where interaction with other agents and road layouts, as well as unexpected behaviors, cannot be fully anticipated and specified ahead of time. Therefore, systems relying on such constraints are likely to silently fail when encountering new situations and tend to be overly conservative when operating in such settings and hence suffer poor performance. The language-based safety critic provides a more dynamic and contextually driven approach to evaluating actions and is more suited to operating in complex settings where safety needs to be generalized beyond specified situations and is also more interpretable to human users.

In this proposed framework, the large language model is used as a risk filter, which estimates candidate actions before their interaction in the simulated environment. Instead of managing the vehicle itself, the critic estimates the likelihood and safety of an action in the context of traffic situation and classify it based on risk level. This pre-execution assessment reduces the risk of executing unsafe actions by filtering high-risk actions at the decision level instead of waiting for their outcomes. The re-planning function also enhances this characteristic in the proposed framework by allowing the policy to make new suggestions in case it determines an action is not safe or is risky. All suggestions are assessed independently by the critic, which enhances the overall likelihood of the selected action being safe. However, in order to make it feasible in real-time systems, there is an explicit limit on re-planning attempts. Therefore, an upper bound is set on re-planning to balance computational resources and time constraints, supporting an effective and robust autonomous driving systems.

Another safety advantage that comes with the *Explain-Then-Act* framework is found in its ability to mitigate failure and prevent catastrophic single-step errors. This is because in a traditional

reinforcement learning pipeline, the actions drawn directly from the policy are acted out immediately. This implies that a single-step error can potentially cause catastrophic outcomes such as collisions or near-misses. The addition of a critique component within the proposed framework adds a cognitive buffer between the policy output and environment interaction. This buffer allows the framework to take a step back and consider the implications of a chosen action before it is carried out. This prevents catastrophic outcomes that may arise as a consequence of a single-step mistake in a traditional reinforcement learning pipeline. Even when the policy produces a suboptimal or unsafe action, the combined design of the safety critic and re-planning limits the impact of the error to a delayed or modified decision rather than immediate execution. As a result, the *Explain-Then-Act* framework improves robustness by converting potentially severe failures into manageable decision revisions within a single time step.

4 Simulation and Performance Evaluation

4.1 Implementation Details

4.1.1 Simulation Environment

The experiments are carried out in the **Highway-env** simulation framework, which is built on the OpenAI Gymnasium interface (Leurent, 2018). **Highway-env** environment is widely used for [RL](#) studies, particularly when autonomous driving behaviors are trained. This provides multi-lane highway with different driving scenarios such as Highway, Merge (with access ramp), Roundabout, Parking (with multiple parking slots and the goal is to park the ego-vehicle with the appropriate heading) and Intersection. This simulation environment allows configuration of the surrounding vehicle count, which enables the ego-vehicle to encounter a range of driving behaviors such as unexpected braking, lane changes, and speed variations.



Figure 4.1: Simulation Environment - highway-env (Leurent, 2018)

The Highway driving scenario with a four-lane and one directional setup to train the ego-vehicle. As shown in [4.1](#), green color box represents the ego-vehicle, while rest of the light blue color boxes are the surrounding vehicles. This simulation environment provide five key actions (Keep speed, Speed up, Slow down, Lane change right and Lane change left) which exactly align with design requirement illustrated in [Figure 3.3](#). Several default values are modified to create a dynamic and competitive environment. The configuration setup is listed as follows [4.1](#)

Table 4.1: Highway-env Configuration Parameters

Parameter	Description	Value
lanes_count	Number of lanes allowed in the environment	4
duration	Maximum episode length in simulation steps	40
policy_frequency	Number of control decisions per second	15
collision_reward	Penalty imposed when a collision occurs	-50
reward_speed_range	Environment uses internal default speed-reward settings	[20,30]
lane_change_reward	Reward given to penetrate lane changes	1.5
high_speed_reward	Reward given to penetrate speed	0.2

4.1.2 Large Language Model

As discussed in 3.2.1, the proposed *Explain-Then-Act* framework includes an LLM that serves as both a critic and a self-explanation module. To balance computational demands, initially a lightweight LLM is selected. The lightweight model of Meta Llama 3.2 (3B) is used for the initial evaluation, which contains 3 billion parameters and outperforms the Gemma 2 and Phi 3.2-mini models. In addition, Meta Llama 3.1 (8B) is used due to its reported good performance in real-world scenarios and reasoning.

Table 4.2: LLM Model Evaluations

Benchmark	Llama 3.2 3B	Llama 3.1 8B
Parameters	3 Billion	8 Billion
<i>General</i>		
MMLU	63.4	73.0
MMLU PRO	—	48.3
IFEval	59.5	80.4
<i>Math</i>		
GSM8K	77.7	84.5
MATH	48.0	51.9
<i>Reasoning</i>		
ARC Challenge	78.6	83.4
GPQA	32.8	32.8

4.1.3 RL Algorithm

As explained in Section 3.2.4, Proximal Policy Optimization (PPO) from Stable Baselines is used to evaluate the *Explain-Then-Act* framework. Several hyperparameters are modified, and Table 4.3 provides a summary of these changes.

Table 4.3: PPO Hyperparameter used in the experiment setup

Parameter	Value
learning rate	0.0003 (default)
batch size	64 (default)
ent_coef	0.015
gae_lambda	0.95
discount factor (gamma)	0.99

Model has been trained for 1 million steps (≈ 2500 episodes) on the HPC Barnard cluster (CPU Cluster) and equipped with 2 x Intel Xeon Platinum 8470 (52 cores) @ 2.00 GHz, further with 512 GB RAM (8 x 32 GB DDR5-4800 MT/s per socket) (NHR Center at TU Dresden, 2025).

4.2 Evaluation Metrics

This section discusses the evaluation metrics developed for the *Explain-Then-Act* framework. The proposed metrics are intended to characterize safety outcomes, assess the impact of policy adjustments, monitor the frequency of LLM interventions, and evaluate the overall performance of the proposed framework.

4.2.1 LLM Intervention Rate

The LLM intervention rate describes how often the LLM identifies a proposed action as unsafe or risky, which requires the agent to generate a new action. A higher percentage value indicates that the proposed actions are frequently classified as unsafe. Further, it enables tracking how the rate changes over the episodes. A decline curve indicates, the agent gradually learn to action safely with fewer unsafe actions. Let R_i be the number of re-planning events in episode i , S be the total number of episode generated during training. Therefore LLM intervention rate I_{LLM} can be defined as

$$I_{LLM} = \left(\frac{1}{S} \sum_{i=1}^N R_i \right) \times 100$$

4.2.2 Average Re-plans per Episode

Average Re-plans per Episode describes the number of instances in which the LLM requests a policy correction due to an unsafe classification within each episode. A higher value indicates

that the policy is repeatedly classified as unsafe in the same episode, while a lower value reflects internal stability and a shift toward safer classifications. Let N be the number of episodes and R_i be the number of re-plans in episode i , therefore the average re-plans can be expressed as

$$\bar{R} = \frac{1}{N} \sum_{i=1}^N R_i$$

4.2.3 Safe Action Adoption Rate

The safe action adoption rate measures how often proposed actions classified as unsafe or risky are corrected within the defined number of re-plans. This metric provides a quantitative view of the ability of the *Explain-Then-Act* framework to mitigate errors from the [RL](#) policy. A higher value indicates that the framework corrects unsafe behaviors effectively, while a lower value shows that unsafe states persist even after go through the defined re-planning attempts.

Let A_{unsafe} be the number of unsafe and risky classifications, R_s be the number of events resolved through re-planning. The safe action adoption rate R_{SA} can be expressed as

$$R_{SA} = \frac{A_{unsafe}}{R_s} \times 100$$

4.2.4 Episode Success Rate

The episode success rate is defined to capture the number of successful episodes. A successful episode means the agent reaches the end goal without any collision or termination due to a time limit. A higher success rate indicates that the *Explain-Then-Act* framework is capable of navigating the [AV](#) without a crash at any point. Let E_s , E_f be the numbers of successful and failed episodes, respectively. Therefore, the episode success rate S_r can be expressed as

$$S_r = \frac{E_s}{E_s + E_f} \times 100$$

4.3 Results Overview

4.3.1 Performance Evaluation

In order to demonstrate the performance of *Explain-Then-Act*, agent is evaluated against a baseline [PPO](#) model, and the moving average reward is used as the primary benchmark for this comparison. The solid line in the plot indicates the moving average performance, and the shaded regions represent the 95 % confidence interval, highlighting the stability and reliability of the *Explain-Then-Act* framework. The traditional [RL](#) model without the [LLM](#) component is selected as the baseline. Both models are simulated in the same environment, with the same [PPO](#) hyperparameter and equal time steps. The moving average reward is recorded and plotted

as illustrated in Figure 4.2. As illustrated, the *Explain-Then-Act* framework outperforms the baseline and results in a stronger learning curve. Further, the proposed framework accelerates early learning, maintains more consistent gains, and ultimately converges to a higher reward. This proves that by integrating LLM module as safety-critic can improve the learning process relative to the traditional RL models.

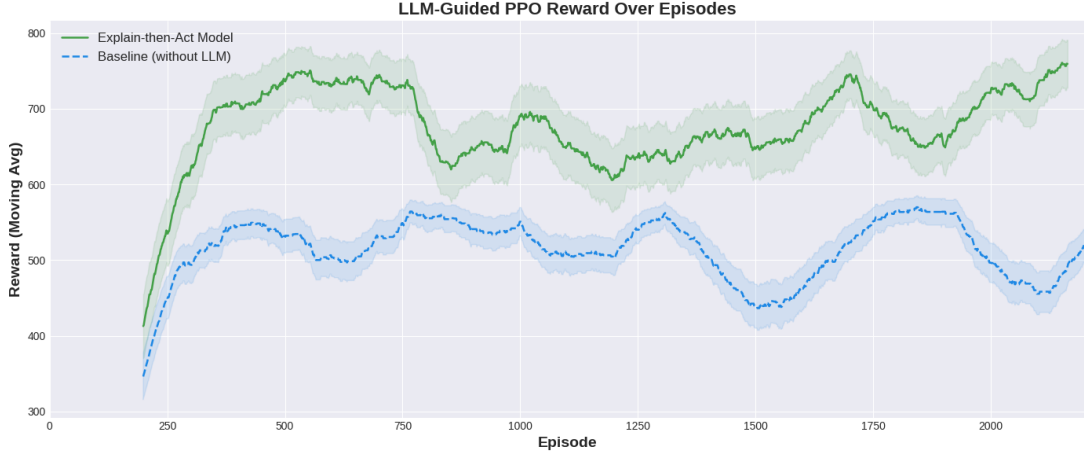


Figure 4.2: Performance Comparison of the Explain-Then-Act Model Against the Baseline Model

Table 4.4: Metric Performance of the Explain-Then-Act Framework

Metric	Values
LLM Intervention Rate	36.69 %
Average Re-plans per Episode	100.81
Safe Action Adoption Rate	83.83 %
Episode Success Rate	79.1 %

As summarized in Table 4.4, the quantitative metric values indicate the performance of the proposed *Explain-Then-Act* framework. In approximately one-third of the timesteps, the LLM influences the agent’s decision-making process, which shows that the LLM provides meaningful interventions in the proposed methodology. On average, nearly 100 re-plans are executed before interaction with the simulated environment. Most importantly, 83.83 % of the actions identified as risky or unsafe are corrected to safer alternatives, highlighting the effectiveness of LLM-guided feedback in improving decision quality. Finally, 79.1 % of the total episodes are completed successfully without collisions or other failures. The baseline model, which does not include the LLM component, records an episode success rate of 65.8 %, while the remaining values are zero since no LLM is involved. Together, these results demonstrate that the proposed framework produces promising outcomes and moves toward a safer and more trustworthy autonomous driving system.

LLM Intervention Rate Changes

The declining LLM intervention rates observed in both the three-re-plan and four-re-plan models indicate that the agent relies less on language-based guidance as training progresses. As shown in Figure 4.3, during the early stages of training, the LLM intervenes frequently because the policy has limited experience and often requires corrective re-planning to address unsafe behaviors in the simulation environment. Over time, the agent gradually learns from this feedback and tends to select safer actions, which reduces LLM involvement. These results further support the goal of this study, which is to show that using an LLM as a safety critic improves the learning process and, over time, enables the policy to act more independently as performance improves.

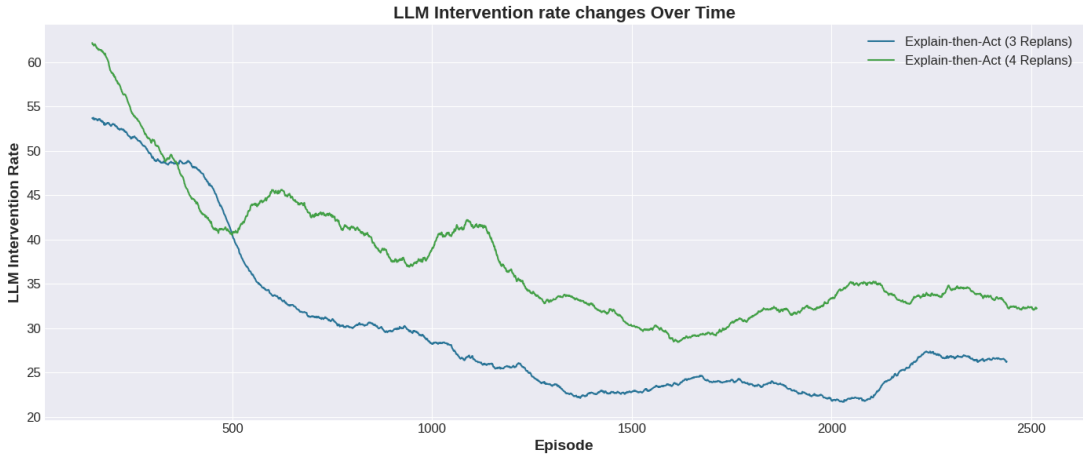


Figure 4.3: LLM intervention rate changes over episodes for the three-re-plan and four-re-plan models

4.3.2 Safety - Efficiency Trade-off Analysis

The introduction of a decision-time safety critique inevitably alters the balance between safety and operational efficiency. In the proposed *Explain-Then-Act* framework, this effect is primarily observed through a slight decrease in average driving speed and an increase in the number of re-planning steps per episode when compared to the baseline reinforcement learning policy. These changes arise because the language model critic evaluates proposed actions against safety-relevant contextual features such as headway distance and time-to-collision before execution. When an action is classified as risky or unsafe, the agent initiates a re-planning cycle, which can delay maneuver execution and favor more conservative alternatives. As a result, the vehicle may maintain its lane longer, reduce speed earlier, or avoid aggressive lane changes that would otherwise increase task efficiency. Although these behaviors may reduce short-term performance metrics related to speed or episode duration, they reflect a deliberate shift toward risk-aware decision making rather than a limitation of the control policy itself.

From a safety standpoint, it is expected and even desired that such a trade-off is made within

the context of autonomous driving systems. Driving efficiency metrics such as travel time or speed optimization are secondary objectives when compared to collision avoidance and compliance with safety constraints. The additional re-planning overhead introduced by the language model critic functions as a safeguard that filters out actions with a higher likelihood of leading to unsafe interactions. This aligns with established principles in safety-critical system design, where conservative behavior is preferred over aggressive optimization under uncertainty. Importantly, the results indicate that the Explain-Then-Act framework achieves this increased safety without causing a collapse in task performance or learning stability. Instead, it demonstrates that a controlled reduction in efficiency can yield a substantial improvement in behavioral robustness. This trade-off supports the argument that decision-time safety reasoning provides a practical and ethically grounded approach for deploying reinforcement learning policies in real-world autonomous driving environments, where regulatory approval and public trust depend on predictable and cautious system behavior.

4.3.3 Further Experiments

1. Impact of Re-planning Frequency on Performance

The *Explain-Then-Act* framework includes a re-planning module, as described in Section 3.2.2. The main purpose of the re-planning module is to trigger the policy to select a different action if the LLM-critic identifies the proposed action as unsafe or risky. The wrapper then selects a new action from the PPO policy and this process continues until the action is classified as safe or the maximum number of allowed re-plans is reached. In this study, three variants are compared based on the number of defined re-plans: two re-plans, three re-plans, and four re-plans.

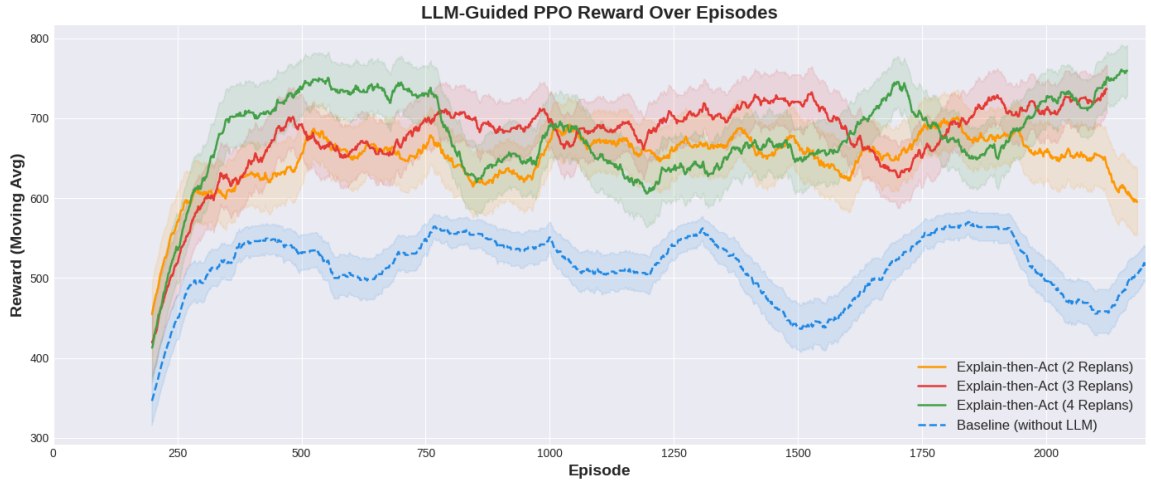


Figure 4.4: Comparison of different re-plans against the baseline

As shown in Figure 4.4, the LLM-guided cumulative PPO reward curves are plotted across episodes for the variants with two, three, and four re-plans. The curves indicate that the number

of allowed re-plans impact on both the learning process and the final return.

The re-planning frequency in the *Explain-Then-Act* framework defines how tightly the policy is coupled with the LLM-critic. When the allowed re-plans are higher, the framework has greater capacity to filter unsafe or risky actions before the agent interacts with the environment. This helps avoid collisions in the early stages of learning and leads to higher initial rewards. In contrast, higher re-plans expose the agent to a larger proportion of safe actions, which leads to two consequences. First, this creates a limited subset of behaviors that are classified as safe and eventually reduces exploration. Second, a higher LLM intervention can make the reward less stationary because the same underlying state may lead to different executed actions depending on how many re-plans occurred in that step. On the other hand, lower frequencies allow fewer corrections, which leaves more unsafe actions in the training process. Although this preserves exploration, it also exposes the agent to more collisions and lower-return episodes in the early stages of training. Both effects are visible in the reward curves, where the trajectories for different re-planning counts show distinct convergence patterns.

Table 4.5: Comparison of evaluation metrics across different re-plans

Metric	2 re-plans	3 re-plans	4 re-plans
LLM Intervention Rate	21.88 %	31.71 %	36.69 %
Average Re-plans per Episode	169.31	149.58	100.81
Safe Action Adoption Rate	63.75 %	76.22 %	83.83 %
Episode Success Rate	40.29 %	57.97 %	79.1 %

Overall, it is essential to maintain a balance between safety and exploration. A lower frequency allows more autonomy for the policy, and the LLM-critic functions only as a light filter. In contrast, higher frequencies, such as four re-plans, strongly filter unsafe or risky actions in the initial proposal before execution.

2. Performance comparison on different LLM Models

In this study, reward curves from different LLM models are compared. Two lightweight LLM models are used: Llama 3.2 (3B parameters) and Llama 3.1 (8B parameters). The main reason for choosing these two models is to balance computational resources and performance. A detailed explanation of each model is provided in Section 4.1.2. As shown in Figure 4.5, *Explain-Then-Act* model with the Llama3.1 (8B) module achieves higher rewards than the one with Llama3.2 (3B) model due it larger capacity and more consistent reasoning patterns suggesting that combining the framework with more sophisticated LLM models leads to stronger results since they are more capable of complex reasoning.

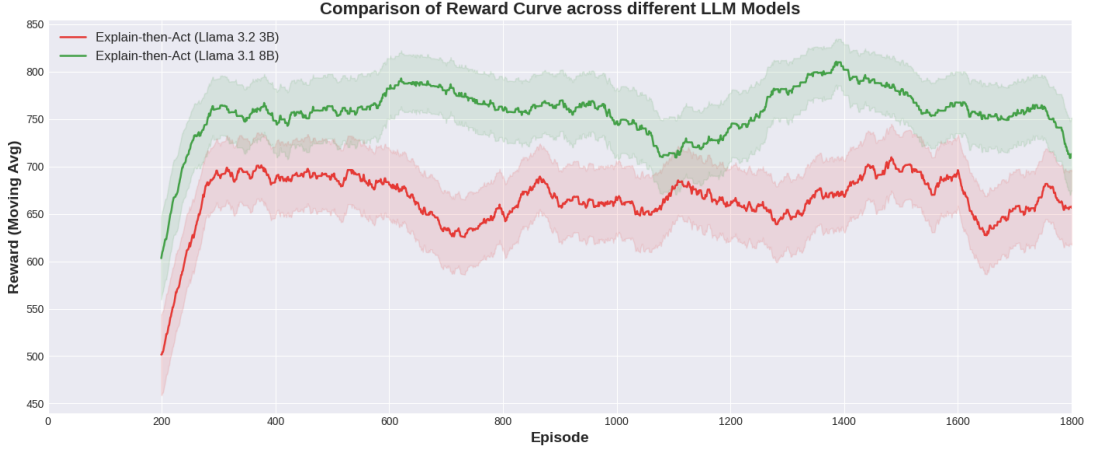


Figure 4.5: Performance Comparison of the Explain-Then-Act Model on different LLM Models

Table 4.6: Comparison of evaluation metrics across different LLM Models

Metric	Llama 3.2 3B	Llama 3.1 8B
LLM Intervention Rate	36.69 %	32.1 %
Average Re-plans per Episode	100.81	78.56
Safe Action Adoption Rate	83.83 %	91.47 %
Episode Success Rate	79.1 %	86.32 %

3. Prompt Engineering Experiment : Impact on Performance and Real-World Evaluation

In this study, the language model evaluates the driving context, identifies potential risks associated with the environment, and provides a safety-oriented classification (safe, risky, or unsafe) that influences the reinforcement learning policy. The LLM provides its interpretation based on the text input, referred to as the prompt. Small adjustments in wording, structure, or level of detail can shift the model’s classification and the weight it assigns to a situation. Loya et al., 2023 describes that even minor variations in prompts can lead to measurable impacts on decision-making and that such systems are sensitive to input text.

For this experiment, the LLM-critique classifications are compared against the MiTra dataset, which is collected from real driving data (Chaudhari et al., 2025). The MiTra dataset closely matches the simulation environment used in this study, as it also represents a highway setting with multiple lanes. In this work, the ramp section of the MiTra dataset is excluded, and the driving actions (as defined in the action space in Section 3.1) are categorized by applying thresholds to longitudinal acceleration and latitude acceleration. Such classification is essential in order to be compatible with the *Explain-Then-Act* framework. Figure 4.6 clearly illustrates how closely the driving action classification aligns with the vehicle speed. Lane changes can be identified using the sign of the lateral acceleration in the dataset. Once the driving actions are

identified, they are combined with the environment context (such as speed, orientation, distance to the front vehicle, and [TTC](#)) and used to construct the prompt as discussed in [Section 3.2.1](#).

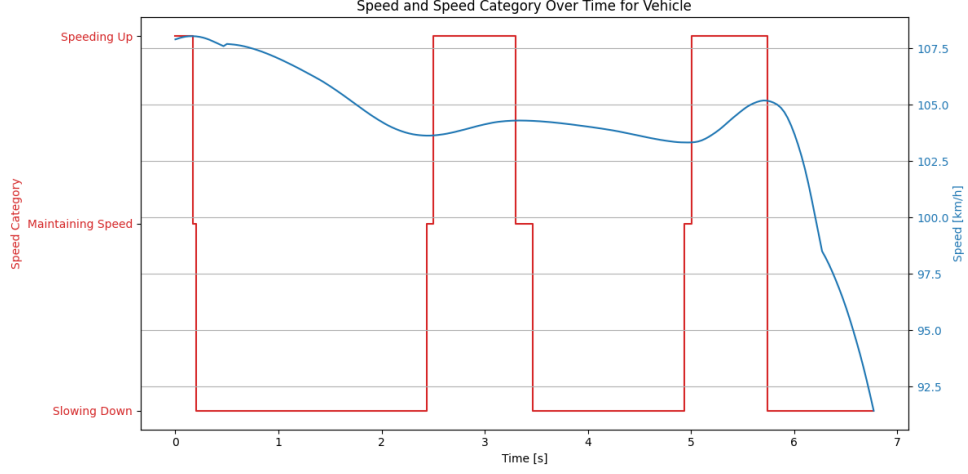


Figure 4.6: Defining the driving actions with MiTra dataset

In the next stage of the study, the classifications generated by the [LLM](#) safety critic are compared across two prompts. As shown in [Figure 4.7](#), the initial prompt classifies the majority of actions as unsafe ([Figure 4.7a](#)). The MiTra dataset did not report any collisions during data collection, and this classification reflects overly cautious behavior, even though the vehicle did not perform any near-miss maneuvers. Therefore, this initial prompt has been adjusted to produce a more balanced classification, so that the LLM-critic flags an action as unsafe only when there is a valid situation. After the prompt change, the majority of actions are classified as risky, as shown in [Figure 4.7b](#).

Two reward curves (before and after the prompt change) are compared for both the three-re-plan and four-re-plan models. As illustrated in [Figure 4.8](#), in both cases, the reward curves after the prompt adjustment outperform. The underlying MiTra dataset provides clear guidance for prompt engineering, and this study supports further strengthening of the framework.

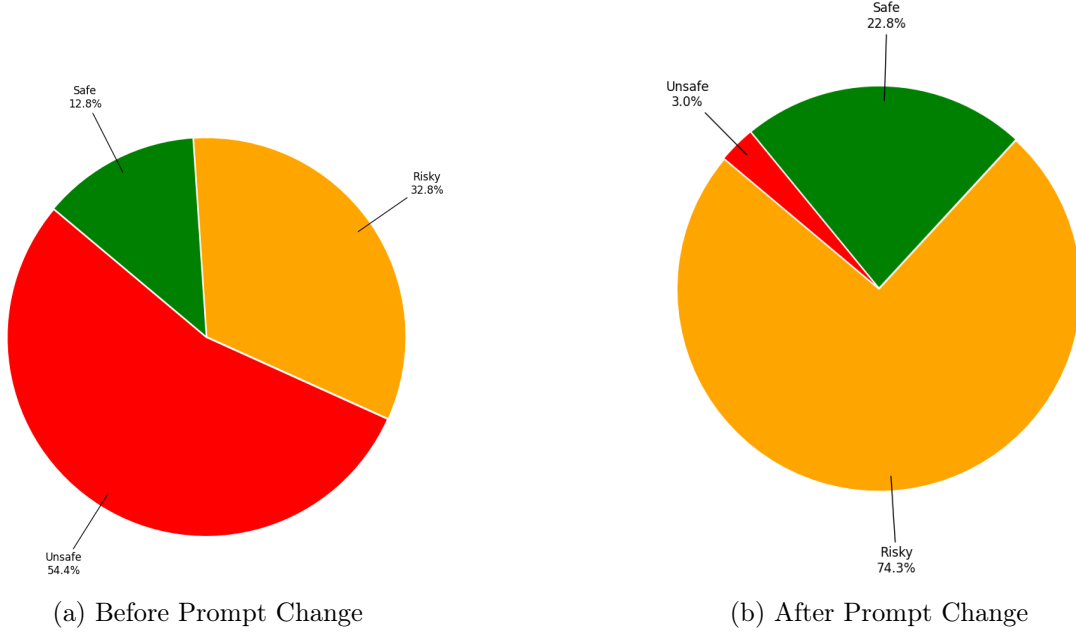


Figure 4.7: Safety-Critic Classification - Before vs After Prompt Changes



Figure 4.8: Reward curves before and after prompt correction - (3 re-plans & 4 re-plans)

4.4 Explainability and Behavioral Analysis

This section explores the effect of the proposed *Explain-Then-Act* framework on the interpretability and behavioral properties of the self-driving agent. Instead of focusing on the quantitative performance measures, this analysis explores the role of decision explanations provided by the large language model in explaining the behaviors of the agent, the detection of safety-related patterns, and the encouragement of transparency in reinforcement learning-based autonomous driving systems. This section clarifies the differences between the baseline reinforcement learning agent and the *Explain-Then-Act* agent, especially in the context of action explanations, safety actions, and their associated trust and inspectability.

As shown in Figure 4.9, 4.10, 4.11 with safety feedback are provided based on the driving context and the chosen action, aligned with traffic rules and environmental constraints such as speed limits.

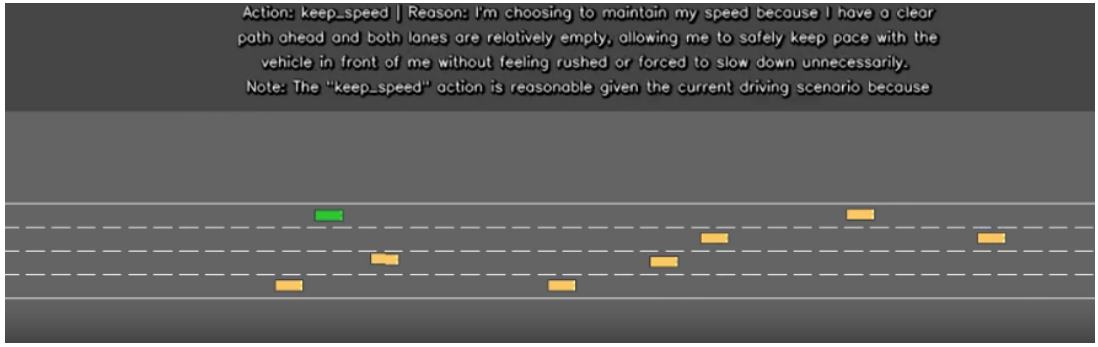


Figure 4.9: Scenario I : Chosen Action - Keep Speed

Explanation: Choose to maintain my seed because I have a clear path ahead and both lanes are relatively empty, allowing me to safely keep pace with the vehicle in front of me without feeling rushed or forced to slow down unnecessarily.

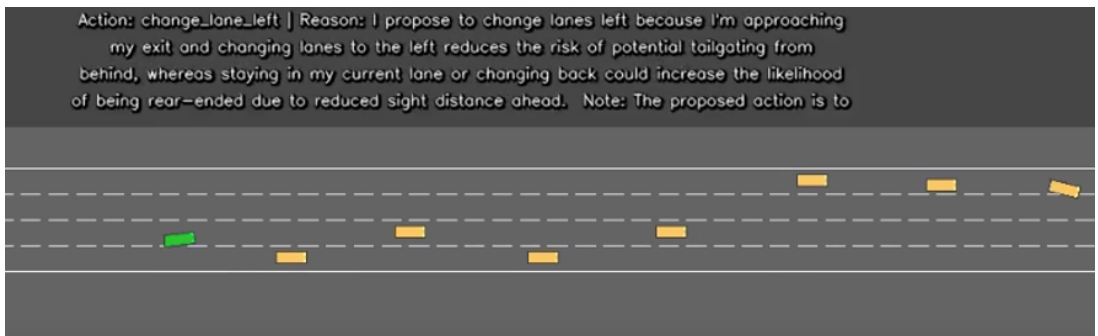


Figure 4.10: Scenario II : Chosen Action - Change Lane Left

Explanation: Choose to change lane left because I'm approaching my exit and changing lanes to the left reduces the risk of potential tailgating from behind, whereas staying in my current lane or changing back could increase the likelihood of being rear-ended due to reduced sight distance ahead.

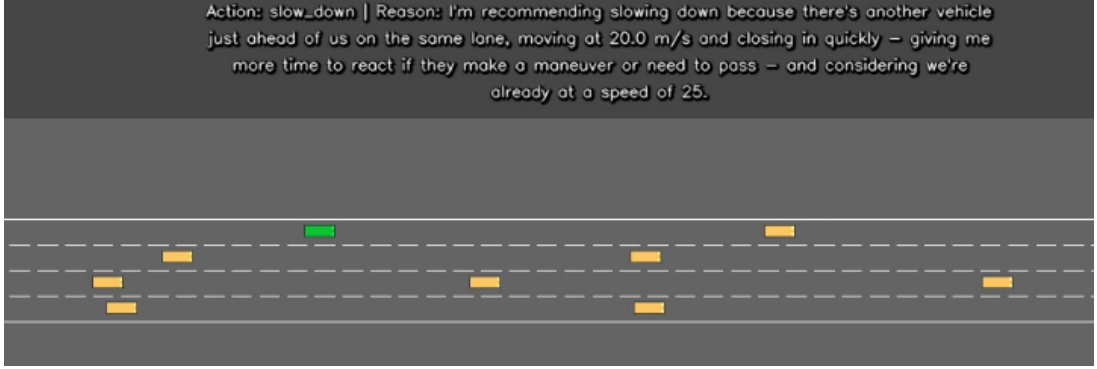


Figure 4.11: Scenario III : Chosen Action - Slow Down

Explanation: Choose to slow down because another vehicle just ahead of u on the same lane, moving at 20.0 m/s and closing quickly. We are already at the speed of 25 m/s

4.4.1 Action-Level Explainability of Driving Decisions

In classical reinforcement learning-based autonomous driving systems, actions such as changing lanes or changing speed are executed without any explicit reasoning or justification. While such systems are capable of achieving high levels of performance, these systems lack of explicit reasoning makes it difficult to understand why an action was executed in a particular situation in a certain traffic setting. The *Explain-Then-Act* approach solves this problem in autonomous driving systems in the following way.

For each proposed action, the safety critic evaluates the context and provide a qualitative assessment that refers interpretable driving variables such as relative speed, headway distance, time-to-collision, and lane occupancy. As a result, each of the agent's decisions, such as slowing down or changing lanes, is justified based on the relationship between the action and the environmental constraints. Such explanations are not possible with the baseline model because of the lack of connection between the agent's decisions and its surrounding.

4.4.2 Behavioral Differences Between the Baseline and Explain-Then-Act Agents

Beyond explainability, the introduction of the LLM-based safety critic leads to observable changes in driving behavior. Compared to the baseline reinforcement learning agent, the Explain-Then-Act agent demonstrates more cautious and anticipatory behavior, particularly in dense traffic scenarios. Unsafe lane-change attempts and aggressive acceleration actions proposed by the policy are frequently revised following safety critique, which results in smoother and more conservative driving patterns.

The re-planning process plays a crucial role in enabling these differences in behavior. Upon identifying that the action is either risky or unsafe, the agent is encouraged to reconsider its

choice instead of taking immediate action. This has a direct effect of reducing the chances of sudden maneuvers while encouraging more safe alternatives that better take into consideration other vehicles on the road. The policy learns from this additional information over time, leading to a decrease in the number of high-risk decisions being made. However, it is crucial to note that the *Explain-Then-Act* framework does not override the reinforcement learning policy. The framework acts as a filter that restricts unsafe behaviors but ensures that the policy is able to operate efficiently. This distinction explains why the performance metrics such as episode success rate remain comparable to the baseline while safety-related behaviors improve.

4.4.3 Implications for Trust, Transparency, and System Inspection

The availability of natural language explanations has important implications for trust and transparency in future autonomous driving. For the passenger, natural language explanations provide comfort in that they clarify why an autonomous vehicle chooses to act cautiously in certain situations. For engineers, such explanations make it easier to debug systems by realizing what environmental factors influenced a particular choice.

The *Explain-Then-Act* approach also facilitates analysis on the episode level. Engineers can analyze the explanations generated during critical episodes for systematic issues with these explanations being overly conservative or overly permissive in safety assessment. Since these explanations are generated without modifying the actual values of the policy parameters directly, modifications can be made on the level of the prompt design or safety thresholds rather than retraining the reinforcement learning model.

However, the presence of explainability in the framework makes way to understand the decision-making process. . The quality of explanations depends on prompt construction and the accuracy of the contextual information provided to the LLM. Nevertheless, the framework represents a substantial improvement over black-box reinforcement learning approaches by making safety reasoning observable and interpretable. This capability is particularly valuable in safety-critical and regulated domains such as autonomous driving, where accountability and transparency are essential.

5 Conclusion and Future Research Directions

Autonomous driving systems often face challenges related to safety and reliability due to the opaque nature of their planning and decision-making components. Such ADS mainly rely on deep reinforcement learning, which provides strong performance even in complex environments, yet the decision-making process remains hidden and is not directly observable or interpretable. It limits the ability to understand internal mechanisms, particularly in safety-critical situations such as accidents or unusual behavior on the road. Explainability at the time of decision-making is therefore important for all parties to build a trustworthy system that ensures consistent autonomous vehicle behavior and alignment with safety regulations. Therefore, the *Explain-Then-Act* framework is introduced to improve explainability by integrating a Large Language Model (LLM) as a safety critic within the reinforcement learning pipeline. This design enables pre-action introspection, supports human-understandable justifications, and provides safety-oriented feedback without compromising control stability.

This thesis makes several original contributions to the field of autonomous driving and explainable reinforcement learning. First, it proposes the *Explain-Then-Act* framework, which augments a standard reinforcement learning policy with a pre-execution safety critique mechanism, natural language explanations, and an iterative re-planning process guided by a large language model. Within this framework, a language model is integrated as an explicit safety critic that evaluates candidate actions before execution, classifies them as Safe, Risky, or Unsafe, and provides human-readable justifications based on the driving context. In addition, the work introduces a hybrid reward function that combines the conventional environment-based reward with a safety-aware feedback signal derived from the language model’s critique, allowing safety considerations to directly influence policy optimization. A key conceptual contribution lies in demonstrating that explainability can be embedded within the decision-making loop itself rather than being limited to post hoc analysis after actions have been taken. Finally, through systematic simulation experiments, the thesis empirically validates that the proposed approach leads to improved safety performance, higher-quality action selection, and increased interpretability of agent behavior when compared to a baseline reinforcement learning policy.

The experimental results show the effectiveness of the proposed *Explain-Then-Act* framework in improving safety outcomes without degrading learning efficiency. In all the test scenarios, the model showed a better ability to take safe actions and a higher success rate compared to the baseline [PPO](#) policy, while significantly reducing unsafe behaviors. These findings imply the positive effect of the language-based safety critic in improving the quality of the safety-critical decisions. The LLM-driven re-planning module caused a significant effect on the model’s behavior by making it considerate and conservative in its decisions based on the surrounding environment. The re-plan ablation studies showed the effectiveness of allowing a certain number of re-planning attempts to reach a balance point where safety and efficiency are balanced, and allowing a few re-plans may hamper the effectiveness, while allowing more re-plans may cause a significant increase in computational time without providing a proportional benefit. The comparison among the different [LLM](#) models showed the effectiveness of the model with better reasoning performance in providing consistent safety judgment, while all the models outperformed the baseline in improving the safety outcomes.

The proposed *Explain-Then-Act* framework improves explainability in autonomous driving by embedding reasoning directly into the decision-making process, which marks a clear departure from both classical deep reinforcement learning and post-hoc explainable reinforcement learning approaches. In contrast to classical [DRL](#) policies that output actions without exposing underlying reasoning, the framework introduces an explicit critique stage in which each candidate action is evaluated and justified before execution. Compared to post-hoc [XRL](#) methods that attempt to reconstruct explanations after actions have already been taken, the proposed approach provides explanations at decision time, which supports timely debugging and meaningful auditing of system behavior. The use of natural language explanations enables engineers and safety analysts to trace why specific actions were accepted or rejected under given traffic conditions, while action-level reasoning makes the agent’s behavior accessible to non-expert stakeholders. The large language model operates as an internal self-reflection mechanism by assessing planned actions against contextual safety constraints and generating qualitative feedback, thereby addressing a key gap in the literature related to the absence of introspection and self-critique in reinforcement learning agents. By making decision rationales observable and interpretable, the framework contributes to increased public trust, supports regulatory acceptance through transparent safety reasoning, and facilitates more effective human–AI collaboration in the development and evaluation of autonomous driving systems.

The proposed approach presents several limitations that should be considered when interpreting the results. First, the effectiveness of the safety critique mechanism depends heavily on prompt quality and on the consistency and correctness of the large language model’s responses, which may vary across scenarios and model configurations. Second, the framework introduces additional computational overhead, as multiple re-planning iterations and repeated language model

inference increase decision-time latency compared to standard reinforcement learning pipelines. Third, the experimental evaluation is limited to simulated highway-driving scenarios with a discrete action space, which constrains the generalization ability of the findings to more complex environments such as urban traffic or continuous control settings. Finally, the study does not include validation in real-world driving systems, nor does it provide formal safety guarantees, which remain essential requirements for deployment in safety-critical applications. Acknowledging these limitations clarifies the scope of the contribution and highlights important directions for further investigation.

The *Explain-Then-Act* framework can also be further improved by future studies. An important extension would be the consideration of continuous action spaces, allowing for greater control over action spaces such as steering and acceleration. Training and testing can also be done under more realistic settings, such as urban driving scenarios involving traffic lights, merging, intersections, and multi-agent interactions. These will allow for the construction of even better automated driving systems that better model the real world. The safety critique component can also be improved by considering multi-modal inputs that integrate visual features with language reasoning. Prompt design can also be further improved by considering adaptive or learned prompting strategies that dynamically change over the training process rather than fixed designs. Smaller and distilled language models may solve the computational issues concerning latency. Finally, combining the LLM-based critique with formal verification or rule-based safety constraints may provide stronger safety assurances by aligning language-based reasoning with mathematically grounded guarantees.

The work presented in this thesis represents a step toward building a trustworthy, transparent, and safe system that delivers promising performance through a novel methodological framework known as *Explain-Then-Act*. With increased familiarity and capability in large language models, this study positions the LLM as a safety critic rather than a post hoc narrator. This design allows the framework to evaluate actions and identify risky or unsafe behavior before interaction with the environment. It addresses a key limitation in the existing literature, where traditional RL-based methods lack interpretable and self-critical mechanisms in safety-critical settings. The results suggest that integrating LLM with RL in autonomous driving can deliver strong performance while providing greater explainability of the decision-making process. More broadly, the proposed approach highlights a viable direction for next-generation autonomous driving architectures in which learning-based control is complemented by explicit reasoning layers, supporting safer deployment and greater societal acceptance of autonomous driving systems.

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Declaration of authorship

I hereby declare that I prepared the paper at hand autonomously, that I did not use any other sources and tools than those listed and that I did not submit this or a similar version of this paper at a different faculty or school for earning an academic degree. Thoughts taken directly or indirectly from other sources are marked as such.

AI tools were used to correct grammar, spelling, and Latex syntax throughout this thesis. Specifically, OpenAI ChatGPT 5.2 and ChatGPT 5.1 models were employed to support these tasks.

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Thanippuli Appuhamilage Dimuth Indeewara